

KAZAKH-BRITISH TECHNICAL UNIVERSITY

School of Information Technology

**DEVELOPMENT OF AN INTELLIGENT SYSTEM FOR
INVESTIGATING AND SOLVING ECOLOGICAL PROBLEMS**

**РАЗРАБОТКА ИНТЕЛЛЕКТУАЛЬНОЙ СИСТЕМЫ ДЛЯ
ИССЛЕДОВАНИЯ И РЕШЕНИЯ ЭКОЛОГИЧЕСКИХ ПРОБЛЕМ**

Master's Thesis

Programme 7M06106 "Software Engineering"

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Almaty, 2026

0.1. ABSTRACT / АННОТАЦИЯ / АНДАТПА

0.1.1. ABSTRACT (English)

Development of an intelligent system for investigating and solving ecological problems

Master's thesis, Program 7M06106 "Software Engineering", KBTU, 2026
Supervisor: Naizabayeva Lyazat Kydyrgaliyevna, Doctor of Technical Sciences, Professor

The relevance of the present work is determined by the acute public health challenge of wintertime PM_{2.5} air pollution in Almaty, Kazakhstan. The city's mountain-basin topography and coal-dependent private-sector heating combine to produce seasonal inversion episodes in which the atmospheric boundary layer collapses to 10–50 m and daily-mean PM_{2.5} concentrations reach 74–81 µg/m³, five to six times the WHO 24-hour guideline of 15 µg/m³. Despite this severity, no open, reproducible, locally calibrated short-term PM_{2.5} forecast has been available to residents or municipal agencies.

The **aim** of the work is to develop and evaluate an intelligent short-term PM_{2.5} forecasting system for Almaty built entirely on open data and deployable on free-tier infrastructure.

The **object of study** is short-term PM_{2.5} concentration dynamics in Almaty. The **subject of study** is machine-learning models for hourly PM_{2.5} forecasting on low-cost-sensor observations under inversion-dominated mountain-basin conditions.

Research methods include data engineering via the OpenAQ v3 and OpenMeteo APIs, supervised learning with gradient-boosted trees (XGBoost) and Ridge regression, temporal 80/20 train-test splitting, error-regime analysis under varying boundary-layer regimes, and comparison against persistence and CAMS global chemistry-transport model baselines.

The **principal results** are as follows. XGBoost achieves RMSE = 23.44 µg/m³ and R² = 0.545 at the 6-hour forecast horizon on a 2,311-hour held-out winter test set (December 2025 – April 2026), outperforming the CAMS global model by 41% in RMSE. Ridge regression outperforms XGBoost at 12 and 24-hour horizons (RMSE 25.85 and 27.24 µg/m³ respectively), establishing that model complexity must be matched to forecast horizon and training-set size. All locally trained models substantially outperform CAMS (negative R² at all horizons), confirming the value of local LCS-based calibration over freely available global chemistry-transport output.

Scientific novelty: (1) the first publicly evaluated PM_{2.5} forecasting model calibrated for Almaty's LCS network and inversion-dominated winter meteorology; (2) a fully reproducible open-source pipeline directly transferable to other Central Asian cities sharing Almaty's data-availability profile. **Practical significance:** a free-access Streamlit web application providing 6, 12, and 24-hour PM_{2.5} forecasts for Almaty with no authentication requirement.

The thesis consists of **60 pages, 3 figures, 8 tables, 25 sources, 1 appendix.**

Keywords: PM2.5 forecasting, Almaty, XGBoost, low-cost sensors, boundary layer height, air quality, gradient boosting

0.1.2. АНДАТПА (Қазақша)

Экологиялық мәселелерді зерттеу және шешуге арналған интеллектуалды жүйені әзірлеу

Магистрлік диссертация, 7М06106 «Бағдарламалық жасақтама инженериясы» мамандығы, ҚБТУ, 2026 ж.

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Жұмыстың **өзектілігі** Қазақстанның Алматы қаласында қысқы мезгілдегі PM2.5 бөлшектерімен ауа ластануының өткір қоғамдық денсаулық мәселесімен анықталады. Қаланың тау қазаншұңқыры топографиясы мен жеке сектордың көмірмен жылытуы инверсия эпизодтарын туғызады: атмосфера шекаралық қабатының биіктігі 10–50 м-ге дейін төмендейді, ал PM2.5 тәуліктік орташа концентрациясы 74–81 мкг/м³-ге жетеді, бұл ДДҰ нормасынан (15 мкг/м³) бес-алты есе жоғары. Тұрғындар мен қалалық қызметтер үшін ешқандай ашық, қайта шығарылатын, жергілікті калибрленген қысқамерзімді болжам болмаған.

Жұмыстың **мақсаты**, тек ашық деректер көздеріне негізделген және тегін бұлттық инфрақұрылымда орналастырылатын Алматы үшін PM2.5 концентрациясы қысқамерзімді болжайтын интеллектуалды жүйені әзірлеу және бағалау.

Зерттеу объектісі, Алматыдағы PM2.5 концентрациясының қысқамерзімді динамикасы. **Зерттеу пәні**, инверсиялы тау котловинасы жағдайында арзан датчиктер (LCS) деректері негізінде PM2.5 сағаттық болжауына арналған машиналық оқыту модельдері.

Зерттеу әдістері: OpenAQ v3 және Open-Meteo API арқылы деректер инженериясы, XGBoost градиенттік бустинг және жоташа регрессия негізінде бақылаулы оқыту; 80/20 уақыттық бөлу бойынша тексеру; қателік режимдерін талдау; персистентті болжам мен CAMS жаһандық химиялық тасымал моделімен салыстыру.

Негізгі нәтижелер: XGBoost 6 сағаттық горизонтта RMSE = 23,44 мкг/м³ және $R^2 = 0,545$ мәндерін көрсетеді, CAMS-тан RMSE бойынша 41%-ға асып озады. Жоташа регрессия 12 және 24 сағаттық горизонттарда XGBoost-тан жақсырақ нәтиже береді (RMSE тиісінше 25,85 және 27,24 мкг/м³). Барлық жергілікті үйренілген модельдер барлық горизонттарда CAMS-тан едәуір жоғары нәтиже береді ($R^2 \leq -0,18$).

Ғылыми жаңалығы: (1) Алматының LCS желісі мен инверсиялы қыс метеорологиясына калибрленген бірінші жарияланған PM2.5 болжау моделі; (2) Орталық Азияның басқа қалаларына ауыстырылатын толық шығарылатын ашық бағдарламалық желі. **Практикалық маңыздылығы:** аутентификация қажет етпейтін, Алматы үшін 6, 12 және 24 сағаттық PM2.5 болжамдарын ұсынатын ашық Streamlit веб-қосымшасы.

Диссертация 60 беттен, 3 суреттен, 8 кестеден, 25 дереккөзден, 1 қосымшадан тұрады.

Түйін сөздер: PM2.5 болжау, Алматы, XGBoost, арзан датчиктер, шекаралық кабат биіктігі, ауа сапасы, градиенттік бустинг

0.1.3. АННОТАЦИЯ (Русский)

Разработка интеллектуальной системы для исследования и решения экологических проблем

Магистерская диссертация, специальность 7М06106 «Программная инженерия» КБТУ, 2026 г.

Научный руководитель: Найзабаева Лязат Кыдыргалиевна, доктор технических наук, профессор

Актуальность работы определяется острой проблемой загрязнения атмосферы воздуха взвешенными частицами PM2.5 в зимний период в Алматы. Горно-котловинная топография города в сочетании с угольным отоплением частного сектора формирует инверсионные эпизоды, при которых высота пограничного слоя атмосферы снижается до 10–50 м, а суточная средняя концентрация PM2.5 достигает 74–81 мкг/м³, в пять-шесть раз выше норматива ВОЗ (15 мкг/м³). При этом ни одного открытого, воспроизводимого, локально калиброванного краткосрочного прогноза качества воздуха для жителей и городских служб Алматы не существовало.

Цель работы, разработка и оценка интеллектуальной системы краткосрочного прогнозирования концентрации PM2.5 для Алматы на основе исключительно открытых источников данных с развёртыванием на бесплатной облачной инфраструктуре.

Объект исследования, динамика краткосрочных концентраций PM2.5 в атмосфере Алматы. **Предмет исследования**, модели машинного обучения для почасового прогнозирования PM2.5 по данным малозатратных датчиков (LCS) в условиях инверсионной горно-котловинной метеорологии.

Методы исследования: инжиниринг данных через API OpenAQ v3 и Open-Meteo; обучение с учителем на основе градиентного бустинга (XGBoost) и гребневой регрессии; валидация с временным разбиением 80/20; анализ режимов ошибок при различной высоте пограничного слоя; сравнение с моделями персистентности и глобальной химико-транспортной моделью CAMS.

Основные результаты: XGBoost достигает RMSE = 23,44 мкг/м³ и R² = 0,545 на горизонте 6 часов на отложенной тестовой выборке из 2 311 часов (декабрь 2025, апрель 2026), превосходя CAMS на 41% по RMSE. Гребневая регрессия превосходит XGBoost на горизонтах 12 и 24 часа (RMSE 25,85 и 27,24 мкг/м³ соответственно). Все локально обученные модели существенно превосходят CAMS (R² ≤ −0,18 на всех горизонтах), подтверждая ценность локальной калибровки на LCS-данных.

Научная новизна: (1) первая опубликованная модель прогнозирования PM2.5, откалиброванная для сети LCS Алматы и инверсионной зимней метеорологии.

(2) полностью воспроизводимый открытый программный конвейер, переносимый на другие города Центральной Азии. **Практическая значимость:** веб-приложение Streamlit с открытым доступом, обеспечивающее прогнозы PM2.5 на горизонтах 6, 12 и 24 часа для Алматы без необходимости аутентификации.

Диссертация состоит из **60 страниц, 3 рисунков, 8 таблиц, 25 источников, 1 приложения.**

Ключевые слова: прогнозирование PM2.5, Алматы, XGBoost, малозатратные датчики, высота пограничного слоя, качество воздуха, градиентный бустинг

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Introduction

0.2. Introduction

0.2.1. *Motivation and relevance*

Almaty's winter air pollution is one of the most severe urban public-health problems in Central Asia, and its physical origins are well understood. The city sits in a natural mountain basin at an elevation of 700–900 m above sea level, enclosed to the south by the Zailiyskiy Alatau range and open to the steppe to the north. Under anticyclonic winter conditions, a stable temperature inversion forms over the basin, suppressing vertical mixing and trapping combustion products near the surface. During the most intense episodes the atmospheric boundary-layer height (BLH) collapses from a summer-season mean of around 1,000 m to values of 10–50 m, reducing the effective volume available for dilution by a factor of 20 to 100. The primary emission source during these episodes is private-sector coal combustion: the great majority of residential heating in peri-urban Almaty relies on solid-fuel boilers, which emit PM_{2.5} and black carbon at rates that are highly sensitive to combustion temperature and coal quality. The combined effect of near-zero ventilation and high emission rates produces daily-mean PM_{2.5} concentrations that routinely exceed 100 µg/m³ at peak winter, placing the city among the most polluted in the IQAir annual rankings for multiple consecutive years (IQAir, 2025). The WHO 24-hour PM_{2.5} guideline is 15 µg/m³ (World Health Organization, 2021); Almaty regularly exceeds this threshold by a factor of six or more during inversion events, and monthly means of 80.7 µg/m³ (December 2025) and 74.2 µg/m³ (January 2026) were recorded in the sensor network used in this study.

Until approximately 2023, the public data infrastructure for monitoring this problem was severely constrained. Almaty's official Kazhydromet network operated fewer than 10 reference-grade monitoring stations within a 682 km² urban area, a density that is inadequate for capturing the fine-scale spatial gradients characteristic of basin-trapped pollution. The emergence of community-scale low-cost sensor (LCS) networks has changed this situation substantially. The AirGradient open-hardware platform, combined with the OpenAQ data aggregation API (OpenAQ, 2024), has enabled a rapid proliferation of low-cost monitors across Almaty. As of April 2026, the network used in the present study comprised 192 sensors within a 25 km radius of the city centre (43.24°N, 76.89°E), of which 153 were AirGradient units concentrated in central and southern residential districts. This density, one sensor per roughly 3.5 km² on average, with higher density in the most densely populated zones, provides a city-scale spatial picture that was impossible to obtain from the reference network alone. Other Central Asian cities facing analogous inversion-driven pollution problems, including Shymkent (Kazakhstan), Karaganda (Kazakhstan), Bishkek (Kyrgyzstan), and Tashkent (Uzbekistan), share the same broad data-availability profile: sparse official monitoring, a growing LCS presence on OpenAQ, and essentially no published machine-learning forecasting work tailored to their specific geography.

Short-term PM_{2.5} forecasting has matured considerably in the European and

North American contexts where reference-grade training targets and dense meteorological networks are available. Gradient-boosted tree models and long short-term memory networks trained on ground-based or satellite-derived features routinely achieve useful skill at 6–24 h horizons. Almaty sits outside this literature for two distinct reasons. The first is data access: the global platforms that publish PM_{2.5} forecasts, IQAir, AirVisual, BreezoMeter, operate proprietary pipelines whose model architectures, training data, and evaluation protocols are not in the public domain. The second is institutional capacity: Kazhydromet publishes hourly observations at its monitoring stations but does not provide a public forecast API or machine-readable hazard thresholds. The scientific literature contains no peer-reviewed study that trains a gradient-boosted model on Almaty’s LCS network and evaluates it against a calibrated freely available baseline on a held-out winter test period. This gap is the direct motivation for the present work.

The significance of this work extends beyond the specific case of Almaty. Central Asia as a region carries one of the highest per-capita PM_{2.5} disease burdens globally, driven by a combination of arid dust sources, coal-heavy energy systems, and rapidly growing cities. A reproducible pipeline that depends only on an OpenAQ API key, a free Open-Meteo weather archive endpoint (Zippenfenig, 2023), and standard Python libraries deployable on free-tier cloud infrastructure represents a transferable template for any LCS-served city in the region. The parametric design of the pipeline - specifically, the fact that city centre coordinates, API query windows, and training configuration can be changed in a single configuration file, means that researchers or municipal agencies in Bishkek, Shymkent, or Tashkent could adapt the system to their local network within hours rather than months, without requiring proprietary data or institutional data-sharing agreements.

0.2.2. Problem statement

The present research addresses the following gap: there is no open, reproducible short-term PM_{2.5} forecasting system built on the LCS network that exists in Almaty, and no published evaluation of how a locally trained model compares to the freely available CAMS global forecast for this specific geography. The scientific gap is the absence of empirical evidence quantifying how much predictive skill is gained by training a machine-learning model on Almaty’s own sensor data relative to applying an off-the-shelf global chemistry-transport model.

Two subsidiary questions are embedded in this gap. First, given the limited size of the available LCS dataset (approximately 9,242 training hours across 562 days), it is unclear whether a non-linear gradient-boosted model or a regularised linear model provides superior accuracy, the answer has direct implications for model selection in analogous data-scarce urban environments elsewhere in Central Asia. Second, the spatial and temporal representativeness of a city-median derived from an unevenly distributed LCS network has not been quantified for Almaty: during sensor outage periods, the median may reflect fewer than 30 active stations rather than the nominal 192, and the consequences for forecast validity are unknown. Addressing these ques-

tions requires a complete evaluation pipeline operating on real, openly accessible data from Almaty’s actual sensor network rather than on synthetic or reference-grade surrogate data. The present study provides that pipeline and reports the first quantitative answers to both questions on a 562-day window ending April 2026.

0.2.3. Scope of the intelligent system

The present thesis realises the general programme described in its title - *development of an intelligent system for investigating and solving ecological problems*, through a focused, high-impact instantiation: wintertime PM2.5 forecasting in Almaty, Kazakhstan. The choice of PM2.5 in an inversion-dominated mountain basin is not arbitrary. It represents the most acute ecological health hazard in the region, the most severe data gap relative to the harm it causes, and the most tractable starting point for an open, transferable intelligent system. The architecture developed here, automated multi-source ingestion, BLH-aware feature engineering, multi-horizon supervised models, and a zero-authentication public forecast interface, is explicitly designed to generalise across ecological monitoring tasks. The same pipeline can ingest NO₂ or O₃ rather than PM2.5; the same coordinate-configurable query layer can serve Bishkek or Tashkent rather than Almaty. The thesis thus delivers both a working system for a specific ecological problem and a reusable template for the broader class of problems named in its title.

0.2.4. Aim and objectives

Aim. Develop and evaluate an intelligent short-term PM2.5 forecasting system for Almaty that is built entirely on open data and code and is deployable on free-tier infrastructure.

Objectives. 1. Ingest and curate PM2.5 observations (OpenAQ LCS) and meteorological features (Open-Meteo, including boundary-layer height) for the 2024-10-01 → 2026-04-15 window into a reproducible SQLite store. 2. Engineer temporal and meteorological features tailored to Almaty’s winter inversion regime. 3. Train and cross-validate an XGBoost forecaster for 6/12/24 h horizons and compare it against three baselines: persistence, linear regression, and CAMS. 4. Analyse error regimes, in particular, episodes of sudden onset and rapid clearing, and articulate the model’s failure modes. 5. Ship a single-screen Streamlit web application that visualises current observations and the XGBoost forecast on a map of Almaty.

0.2.5. Scientific novelty

The first contribution of the present study is the development of the first publicly released PM2.5 forecasting model calibrated specifically for Almaty’s LCS network and inversion-dominated winter meteorology. Prior work on Almaty air quality is predominantly descriptive (source apportionment, trend analysis), or relies on satellite-derived estimates rather than ground-level sensor data. The present model is trained and evaluated on the 192-station OpenAQ network over a 562-day window ending April 2026, using a feature set that explicitly encodes BLH dynamics and inversion-onset signals not present in earlier local work.

The second contribution is the reproducible open architecture. All inputs are openly licensed (CC BY 4.0, no paywalled data); the full pipeline is a single Python repository with no cloud dependencies. Comparable published systems - including those evaluated by (Ayus et al., 2023; Enebish et al., 2021; Menares et al., 2021), typically rely on proprietary ground truth, restricted API access, or institutional data agreements that prevent reproduction or geographic transfer. The present design explicitly eliminates these barriers, making the pipeline portable to Shymkent, Karaganda, and other Central Asian cities with analogous LCS-on-OpenAQ data profiles.

The third contribution is a methodologically significant result regarding model-complexity trade-offs. Ridge regression outperforms XGBoost at the 12 and 24-hour forecast horizons, by 9.8% in RMSE at +12 h (Ridge: 25.85 vs. XGBoost: 28.67 $\mu\text{g}/\text{m}^3$) and by 13.5% at +24 h (Ridge: 27.24 vs. XGBoost: 31.48 $\mu\text{g}/\text{m}^3$). This result is observed on a training set of 9,242 hours with 27 features, a scale at which, on the basis of published benchmarks, non-linear ensemble methods would normally be expected to retain their advantage. The finding contributes to the evidence base on model-complexity trade-offs for air quality forecasting problems with under 10,000 training samples. The most closely related published comparisons (Enebish et al., 2021; Menares et al., 2021) report similar model-size reversals in settings where the target variable carries substantial measurement noise, which is consistent with the known hygroscopic overestimation bias in uncalibrated LCS readings.

This reversal also has an immediate practical consequence for system design. Because Ridge regression requires no tree construction, its inference time on a single row of features is negligible compared to XGBoost, and its serialised model file is several orders of magnitude smaller. In the context of a free-tier Streamlit deployment where memory and compute are constrained, a Ridge model at the 12 and 24-hour horizons is not merely equally accurate but computationally preferable. This alignment of statistical and engineering considerations strengthens the case for a multi-model operational system over a single-model one.

0.2.6. Object, subject, and methods

- **Object of study:** short-term PM_{2.5} concentration dynamics in Almaty.
- **Subject of study:** machine-learning models for hourly PM_{2.5} forecasting on LCS-based observations in inversion-dominated mountain-basin conditions.
- **Methods:** data engineering on openly available APIs; supervised learning with gradient-boosted trees; time-series cross-validation; baseline comparison; error-regime analysis.

0.2.7. Practical significance

The locally trained XGBoost model achieves RMSE = 23.44 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.545$ at the 6-hour forecast horizon on a held-out winter test set of 2,311 hours (December 2025 – April 2026), outperforming the CAMS global chemistry-transport model by 41.1% in RMSE at the same horizon. At the 12-hour horizon, Ridge regression produces RMSE = 25.85 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.446$, outperforming CAMS by approximately 28% in RMSE. At 24 hours, Ridge achieves RMSE = 27.24 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.383$,

while CAMS produces negative R^2 values at all three horizons (-0.314 , -0.376 , and -0.184 at 6, 12, and 24 h respectively), indicating that the global model provides no useful predictive signal for this geography. All locally trained models outperform CAMS by at least 16% in RMSE across every horizon tested.

The practical deployment of these results takes the form of a Streamlit web application that provides a continuously updated PM2.5 forecast for Almaty at no cost to the end user. The application refreshes on a 30–60 second cycle, requires no authentication, and can be accessed from any device with a browser, making it suitable for use by schools, hospitals, municipal public health offices, and community organisations that lack technical staff or data infrastructure. The forecast uncertainty is communicated as a confidence band of approximately $\pm 17 \mu\text{g}/\text{m}^3$ at the 6-hour horizon (± 1 MAE), which provides a directly operationally useful range: an advisory issued when the lower bound of the forecast band crosses the WHO 24-hour guideline of $15 \mu\text{g}/\text{m}^3$ will capture genuine deterioration events at a false-positive rate acceptable for public communication purposes. The map view displays the current city-median PM2.5 alongside the three horizon forecasts, and the colour scale is calibrated to the Kazakhstani national AQI categories so that non-specialist users can interpret the displayed values without reference to $\mu\text{g}/\text{m}^3$ numerics. The complete pipeline, data ingest, feature engineering, model training, and application, is implemented in a single open Python repository with no cloud dependencies, making it directly portable to other Central Asian cities.

Deployment URL: [to be added after Week 3 deployment]

0.2.8. Structure of the thesis

Chapter 1 reviews the scientific and policy context in three parts: the meteorological and emission drivers of Almaty’s PM2.5 episodes, including the role of temperature inversions and coal combustion; the field of low-cost sensor technology and the OpenAQ data ecosystem that makes the present dataset possible; and the methodological literature on gradient-boosted tree models and alternative approaches for short-term urban PM2.5 forecasting, with emphasis on studies conducted in inversion-dominated mountain or basin environments. Chapter 2 describes the data sources and system architecture in full reproducible detail: the SQLite database schema, the OpenAQ and Open-Meteo API ingestion procedures, the 27-feature matrix construction (including BLH delta, temporal lag, and calendar features), the train/test split, and the rationale for each non-obvious design choice. Chapter 3 reports the experimental results: the quantitative model comparison across all three forecast horizons and all four models, the feature importance analysis, the error-regime case studies on inversion onset and clearing events, and the spatial analysis of sensor-coverage effects during the October–November 2025 AirGradient firmware outage period. Chapter 4 discusses the implications of the results against the stated research questions, situates the findings within the published literature, quantifies the principal limitations, and proposes five specific directions for future work, including SHAP-based feature

attribution, forecast BLH integration, and an ensemble of XGBoost and Ridge regression. An appendix provides the full model hyperparameter configurations, the API query specifications, and the data-quality filter thresholds applied during ingestion, so that the complete pipeline can be reproduced from the archived code and database without ambiguity.

1. Analytical Review

1.1. 1. ANALYTICAL REVIEW

1.1.1. *1.1. Air quality in Almaty and Kazakhstan*

1.1.1.1 1.1.1. Topographic and meteorological drivers

Almaty occupies a basin at the northern foothills of the Trans-Ili Alatau mountain range (43.24°N, 76.89°E, elevation 700–900 m above sea level), with peaks rising to over 4,000 m immediately to the south and south-east. This asymmetric terrain configuration is the primary determinant of the city’s air quality regime, and it operates through two interacting mechanisms: cold-air drainage and thermally forced boundary-layer collapse.

Cold-air drainage occurs because the dense, chilled air that accumulates overnight on the mountain slopes flows downslope by gravity and pools in the Almaty basin. The surrounding terrain to the north, where the city transitions to the Kazakhstani steppe, offers relatively unobstructed outflow, but the valley geometry slows horizontal ventilation considerably. The result is that cold surface-air masses can persist for 12–36 hours after the initial nocturnal drainage event, particularly when synoptic-scale anticyclonic conditions suppress vertical mixing over the entire region. During sustained inversion events recorded in the training dataset (2024-10-01 to 2026-04-15), boundary-layer heights (BLH) fall to as low as 10 m, against a summer mean of approximately 1,000–1,500 m and a training-window mean of 257 m. This extreme range, 10 m at the low end, 3,935 m at the peak of thermally active summer afternoons, is itself a quantitative signature of the inversion-dominated climate and is directly exploited by the BLH feature in the forecasting model.

Temperature inversions arise when the vertical temperature profile of the atmosphere is inverted relative to the standard lapse rate: the air aloft is warmer than the surface layer. Under such conditions, vertical convective mixing is suppressed because a parcel of surface air, if displaced upward, encounters air that is warmer and thus more buoyant, and it sinks back. In Almaty, radiation inversions develop on clear winter nights when the snow-covered surface loses heat rapidly through longwave emission, chilling the near-surface air. Surface-based inversions of 5–15 °C have been documented in regional meteorological records (Tursumbayeva et al., 2022). Elevated inversions also form when warm air advects from the south over the mountain range and overrides the cold pool in the basin. Both inversion types act as a lid that confines surface-emitted aerosols to the lower 50–200 m of the atmosphere, where the city’s population lives and breathes.

The seasonal modulation of boundary-layer depth explains why Almaty’s PM_{2.5} problem is almost entirely a winter phenomenon. In summer, strong solar heating destroys the nocturnal inversion by mid-morning and drives the convective boundary layer to 1,500–3,000 m, diluting surface emissions into a volume that is ten to twenty times larger than the winter surface layer. This dilution effect, independent of changes in emissions, is sufficient to reduce ambient PM_{2.5} concentrations from

winter highs exceeding $100 \mu\text{g}/\text{m}^3$ to the $12\text{--}17 \mu\text{g}/\text{m}^3$ summer baseline observed in the present dataset. The seasonality of $\text{PM}_{2.5}$ in Almaty is therefore not a pure reflection of emission seasonality; it is the product of emission intensity multiplied by an extremely variable dilution volume.

1.1.1.2 1.1.2. Emission sources and seasonal patterns

Three primary source categories drive Almaty's $\text{PM}_{2.5}$ loadings: residential coal combustion, thermal power generation, and vehicle traffic. Their relative contributions shift with season and time of day, and their combined effect is amplified by the meteorological drivers described in §1.1.1.

Residential coal heating is the dominant wintertime source. Peri-urban and low-income districts on the city periphery, particularly in the eastern and western micro-districts, lack connection to the centralised district heating network. Households in these areas rely on individual coal-fired stoves or boilers, which operate at much lower combustion temperatures than utility-scale furnaces and therefore emit disproportionately high quantities of unburned carbonaceous particles, fine particulate matter, and black carbon. The heating season in Kazakhstan officially runs from approximately 15 October to 31 March, corresponding to the cold-temperature window when residential heating demand is greatest. This six-month window almost exactly coincides with the high- $\text{PM}_{2.5}$ months in the dataset: the monthly mean of $80.7 \mu\text{g}/\text{m}^3$ in December 2025 and $74.2 \mu\text{g}/\text{m}^3$ in January 2026, against a summer baseline of $12\text{--}17 \mu\text{g}/\text{m}^3$, define a seasonal amplitude factor of approximately five. The sharpness of the heating-season transition, visible in the training time series as a rapid concentration increase in mid-October, motivated the inclusion of a binary heating-season indicator (1 from 15 October to 31 March, 0 otherwise) as a categorical feature in the model.

Thermal power generation contributes through Almaty's two combined-heat-and-power plants, CHP-2 (located in the city's western industrial zone) and CHP-3 (in the northern periphery). Both plants burn Karaganda-basin coal and emit $\text{PM}_{2.5}$, sulfur dioxide, and nitrogen oxides from stacks whose effective injection height exceeds the inversion-layer ceiling under calm conditions. During inversion events, emissions that would normally disperse into the mixed layer instead accumulate near-surface after dry deposition and secondary aerosol formation. The plants collectively supply approximately 70% of the city's district heating load (World Bank, 2022), making them non-dispatchable during the heating season.

Vehicle traffic compounds the problem through a different diurnal cycle. Almaty's passenger car ownership rate is among the highest in Central Asia, and a substantial fraction of the fleet is older petrol and diesel vehicles with limited emissions control. Traffic-related $\text{PM}_{2.5}$, primarily brake dust, tyre wear particles, and direct exhaust, peaks during morning (07:00–09:00 local time) and evening (17:00–19:00) commutes. This diurnal pattern overlaps with the weakest boundary-layer mixing of the day, creating a double-peaked hourly concentration signature that is reproduced

in the lag feature structure of the model. The hour-of-day cyclical encoding captures this pattern without requiring explicit source attribution.

A secondary emission category relevant to summertime periods is dust resuspension. The Almaty basin is bordered to the north by semi-arid steppe, and dry southerly winds in spring carry mineral dust from bare soil surfaces. These dust events can briefly elevate PM_{2.5} to 40–60 $\mu\text{g}/\text{m}^3$ even in months outside the heating season, producing short-duration spikes that are visible as outliers in the summer baseline series. Dust events are not the focus of the present model, but they introduce heteroscedastic residuals in the spring transition period (April–May) that affect the RMSE decomposition in §3.4.

1.1.1.3 1.1.3. Health impacts and regulatory context

The health consequences of sustained PM_{2.5} exposure at Almaty’s observed concentrations are quantitatively severe. The World Health Organization’s 2021 air quality guidelines set the 24-hour PM_{2.5} limit at 15 $\mu\text{g}/\text{m}^3$ and the annual mean guideline at 5 $\mu\text{g}/\text{m}^3$ (World Health Organization, 2021). During the heating season of 2025–26, the cross-sensor median PM_{2.5} in the present dataset exceeded the 24-hour guideline on the majority of winter days. At exposures above 35 $\mu\text{g}/\text{m}^3$, the WHO epidemiological evidence base links each 10 $\mu\text{g}/\text{m}^3$ increment in 24-hour PM_{2.5} to increases in all-cause mortality, cardiovascular hospital admissions, and respiratory emergency department visits (World Health Organization Regional Office for Europe, 2013). Almaty’s approximately two million residents are exposed to these concentrations for roughly six months of each year.

Kazakhstan’s regulatory framework for ambient air quality is established by the Environmental Code of the Republic of Kazakhstan, enacted in January 2021 (Экологический кодекс Республики Казахстан, 2021). The Code sets maximum permissible concentrations (MPC) for PM_{2.5}, mandates environmental impact assessment for major emission sources, and establishes a framework for emission trading and environmental permits. The national AI strategy for 2021–2025 identifies environmental monitoring and smart city applications as priority domains for artificial intelligence deployment. Despite this policy context, there is no publicly accessible, near-real-time, locally calibrated short-term PM_{2.5} forecast for Almaty that is openly documented or independently reproducible. The present work directly addresses this deficit.

The gap between regulatory intent and operational reality has several causes. Kazhydromet’s reference monitoring network, while technically credible, is too sparse to provide the neighbourhood-level spatial detail needed for public exposure guidance, and its data publication lag makes real-time decision support impossible (§1.2.1). The commercial forecast services that do exist for Almaty (§1.2.3) operate without published evaluation metrics. Public health interventions, school closures on high-pollution days, voluntary traffic restrictions, biomass burning bans, require actionable 6–24 hour forecasts that are not currently available from an open, auditable

source.

1.1.2. 1.2. Existing monitoring systems

1.1.2.1 1.2.1. Reference-grade monitoring

The formal air quality monitoring network in Almaty is operated by Kazhydromet, the national hydrometeorological and environmental monitoring service of the Republic of Kazakhstan. Reference-grade monitoring stations in this network use β -attenuation monitors (BAM) or tapered-element oscillating microbalance (TEOM) instruments that meet World Meteorological Organization (WMO) performance requirements for PM_{2.5} mass concentration measurement and provide readings traceable to gravimetric standards. Both instrument types are widely accepted internationally as reference-equivalent methods: BAM instruments operate by drawing a particle-laden sample through a filter tape and measuring the attenuation of a beta-radiation beam, while TEOM instruments determine mass accumulation from the resonance frequency shift of a vibrating glass element. Both achieve measurement uncertainties of ± 5 – 10% under typical operating conditions (World Meteorological Organization, 2016).

The Kazhydromet network is, however, sparse relative to the geographic and demographic scale of Almaty. Fewer than ten operational stations are distributed across a city area of approximately 682 km², yielding an average area per station above 68 km². For comparison, the US EPA NAAQS monitoring guidance recommends station spacing on the order of 10–20 km in urban areas with known emission hotspots, while population-exposure studies typically require even finer resolution to capture neighbourhood-scale gradients. At the observed station density, the Kazhydromet network cannot resolve intra-urban spatial gradients, identify emission hotspots, or provide the spatially aggregated city-scale concentration that would serve as a single-number target for a forecasting model.

Equally constraining for the purposes of the present work is the data accessibility limitation. Kazhydromet does not expose a public, machine-readable API for near-real-time hourly PM_{2.5} data. Data are published in periodic bulletins and annual reports, with lags of days to months between measurement and public release. This lag structure makes Kazhydromet data unsuitable as a training target or real-time forecast anchor for an hourly prediction system. The present study therefore does not use Kazhydromet data as a primary input, though a future co-location study comparing the OpenAQ LCS median against Kazhydromet reference readings would substantially strengthen the quantitative interpretation of model results.

1.1.2.2 1.2.2. Low-cost sensor networks

The community low-cost sensor (LCS) network aggregated through the OpenAQ v3 platform substantially densifies the monitoring coverage available to researchers and policymakers in Almaty (OpenAQ, 2024). As of the API snapshot taken on 20 April 2026, 192 active monitoring locations within 25 km of the city centre (43.24°N,

76.89°E) report PM2.5 data through the OpenAQ platform, providing a spatial density roughly twenty times greater than the Kazhydromet reference network.

The dominant contributor to this network is the Almaty Air Initiative, a citizen-science organisation that has deployed AirGradient ONE open-hardware monitors across residential districts since 2024. AirGradient units account for 153 of the 192 stations (79.7%). AirGradient ONE is an open-source, community-built sensor platform based on the Plantower PMS5003 optical particle counter, whose design schematics and firmware are publicly available under an open-hardware licence. The Clarity Node-S, a commercial LCS device manufactured by Clarity Movement Co. (Berkeley, CA), contributes 38 stations (19.8%). A single AirNow-affiliated station provides the remaining 0.5% of the network. All data transmitted through OpenAQ are licenced under CC BY 4.0, permitting unrestricted research use with attribution. This open-licence condition is a prerequisite for the reproducibility objective of the present work.

The OpenAQ platform’s v3 API provides a standardised JSON interface through which sensor readings, metadata, and quality flags can be queried programmatically. The ingest pipeline described in §2.2 issues hourly requests to the `/v3/locations/{id}` endpoint, filters readings by the plausibility criteria defined in §2.3, and assembles the city-wide PM2.5 median used as the model target. The combination of platform standardisation and open licensing makes OpenAQ the only viable data source for a fully reproducible Almaty forecasting pipeline as of the study period.

1.1.2.3 1.2.3. Commercial aggregators and the data gap

Commercial air quality platforms, notably IQAir and the AirKz aggregator, publish AQI dashboards and, in some cases, AI-based short-term forecasts for Almaty. IQAir’s World Air Quality Report (IQAir, 2024) ranked Almaty among the most polluted cities in Central Asia, citing annual mean PM2.5 concentrations that are consistent with the seasonal patterns documented in the present dataset. These commercial platforms play a valuable role in public communication and policy advocacy.

Their limitation for scientific purposes is fundamental: neither IQAir nor AirKz publishes methodology documentation, calibration procedures, model architecture descriptions, evaluation metrics, or the underlying training data for their forecasting products. The IQAir forecast pipeline, in particular, ingests data from a global network of IQAir AirVisual Pro sensors and applies undisclosed correction algorithms whose performance has not been independently peer-reviewed. Without access to methodology, forecast skill cannot be independently assessed. Without access to training data, the published rankings cannot be reproduced. Without published evaluation against held-out data, any claimed accuracy is unverifiable.

This opacity creates a specific gap that the present work addresses: Almaty lacks a published, open, and reproducible short-term PM2.5 forecast whose inputs, processing steps, model architecture, and evaluation results are fully documented. The value of the present system lies not only in its forecast accuracy, though §3.2

demonstrates a 41% RMSE improvement over the CAMS global baseline at the 6-hour horizon, but in the fact that every step of the pipeline, from API ingest to Streamlit deployment, is version-controlled and replicable by any researcher with access to the public OpenAQ and Open-Meteo APIs.

The reproducibility requirement is not merely an academic formality. A forecast system that informs public health decisions must be auditable: a regulator, a public health official, or a peer reviewer must be able to trace any individual forecast back to its training data, feature specification, and model parameters. Closed commercial pipelines cannot meet this standard. The present work addresses it by publishing the complete pipeline source code, the BibTeX-documented data provenance, and a held-out test set evaluation on 2,311 hours of winter 2025–26 data that was excluded from all training and hyperparameter selection steps. Any user of the open-source repository can reproduce the $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ result at the 6-hour horizon from scratch using only freely available data and a standard Python environment.

1.1.3. 1.3. Low-cost sensors: capabilities and limitations

1.1.3.1 1.3.1. Operating principles and cost structure

Low-cost sensors (LCS) for PM_{2.5} measurement exploit optical particle counting through laser-scattering. An aerosol sample is drawn through a sensing chamber where a laser diode, typically operating at 780 nm wavelength, illuminates the particle stream. Individual particles or clusters scatter the laser light at angles and intensities that depend on their size and refractive index; a photodetector positioned at a defined scattering angle converts this optical signal to a count distribution across particle-size bins. The sensor firmware then applies a lognormal conversion model to estimate PM_{2.5} mass concentration from the particle-count data.

The two dominant LCS platforms in the Almaty network exemplify the diversity within this operating principle. The Plantower PMS5003, embedded in the AirGradient ONE, draws a sample flow of approximately 100 mL/min through a 34 mm × 20 mm sensing volume and outputs concentration estimates for PM_{1.0}, PM_{2.5}, and PM₁₀ at one-minute intervals via serial UART protocol. The Sensirion SPS30, an alternative optical counter used in some AirGradient ONE units, adds a laser-particle-interaction cleaning cycle that reduces particle deposition on the sensing optics, extending the calibrated service life from approximately 8,000 to 12,000 operating hours (Sensirion AG, 2023). The Clarity Node-S employs a proprietary laser-scattering geometry with firmware-level humidity correction, though the correction algorithm is not publicly documented.

The economic case for deploying LCS networks rather than reference instruments is straightforward at these cost levels. Reference-grade BAM and TEOM instruments cost USD 20,000–80,000 per unit, plus installation, calibration, and maintenance costs that can approach the capital cost over a five-year operational period. In contrast, an AirGradient ONE sensor, including enclosure, microcontroller, and connectivity, costs approximately USD 100–150 in component form, and the Clarity Node-S retails for approximately USD 300–400. This 100–500× cost differential

makes the 192-station Almaty network economically feasible for a citizen-science organisation, whereas a reference-grade network of comparable density would cost tens of millions of dollars. The trade-off is that LCS measurements lack the metrological traceability of reference instruments and are subject to the biases described in §1.3.2.

1.1.3.2 1.3.2. Humidity and drift biases

The primary known systematic bias for optical LCS in humid environments is hygroscopic growth of aerosol particles. Above approximately 70% relative humidity (RH), water molecules condense onto the surfaces of hygroscopic aerosol constituents, primarily sulfate, nitrate, and organic carbon species, enlarging the apparent particle diameter. Because LCS firmware translates particle diameter to mass concentration using refractive indices and density assumptions calibrated for dry particles, the measured PM_{2.5} is systematically inflated under humid conditions. Published intercomparisons of Plantower-class sensors against reference BAM instruments in urban environments report positive humidity biases of 20–60% at RH > 80%, with the magnitude depending on the hygroscopic fraction of the local aerosol (Di Antonio et al., 2018) (Kosmopoulos et al., 2020). During Almaty’s winter inversion episodes, precisely the high-concentration events of greatest interest for forecasting, ambient RH frequently exceeds 75–85%, meaning that the LCS readings are likely to be positively biased during the most critical measurement periods.

Inter-unit variability is a second quantifiable source of uncertainty. Factory-calibrated units of the same model show concentration offsets of 10–30% under identical controlled-chamber conditions, attributable to tolerances in laser alignment, photodetector response curves, and firmware parameter sets (Popoola et al., 2016). When units are deployed across a city with diverse microclimates, varying heights above ground (some AirGradient units are mounted at 1.5 m, others at 3–5 m on balconies or rooftops), and different local source influences, the inter-unit spread across the 192 Almaty stations during any given hour substantially exceeds the controlled-chamber 10–30% figure. The standard deviation across all active stations in the present dataset can exceed 30 µg/m³ in a single hour during peak winter episodes, confirming that individual-station readings are not representative of city-wide exposure.

A third bias source is temporal drift: the gradual degradation of laser intensity and photodetector sensitivity over months of field deployment. Laboratory studies of Plantower PMS5003 units subjected to accelerated aging report signal attenuation of 10–25% over approximately 8,000–10,000 operating hours (Sayahi et al., 2019). The training window in the present study spans 562 days (2024-10-01 to 2026-04-15), or roughly 13,500 operating hours for sensors that were already deployed before the training window opened. Systematic drift cannot therefore be excluded, particularly for AirGradient ONE units deployed in the inaugural 2024 network expansion. Without periodic reference co-location, the magnitude of drift in the Almaty network is not directly measurable from the available data.

No co-location calibration against reference instruments was performed in the present study. The PM_{2.5} values derived from the OpenAQ network should therefore be understood as relative concentration signals that track true PM_{2.5} variability but may carry systematic biases of $\pm 20\text{--}50\%$ in absolute magnitude during high-RH winter conditions. This limitation is explicitly acknowledged in the scope discussion (§4.2) and does not invalidate the relative accuracy comparisons among models trained and evaluated on the same LCS-derived target series.

1.1.3.3 1.3.3. Mitigation strategies applied in this study

The standard mitigation for inter-unit variability in a multi-sensor LCS network is spatial aggregation: combining readings from multiple co-located or nearby sensors so that random unit-specific offsets cancel in the aggregate. The present study applies this principle at city scale by computing the cross-sectional median PM_{2.5} across all stations with plausible readings in a given hour. The median, rather than the mean, is chosen because the distribution of station readings in any given hour is typically right-skewed, a small number of stations near localised sources (a coal stove, a road intersection) read substantially above the bulk of the network, and those outliers would inflate the mean disproportionately. The median is resistant to such outliers by construction.

Three additional constraints bound the aggregation. A minimum of three contributing stations per hour is required; hours with fewer than three readings are recorded as missing rather than propagated as unreliable estimates. A plausibility filter excludes readings outside the range $0\text{--}500\text{ }\mu\text{g}/\text{m}^3$, rejecting physically implausible values likely caused by sensor malfunction or communication errors. A further check discards readings from any station whose 24-hour running mean deviates by more than three standard deviations from the network median, flagging potentially faulty or improperly calibrated units. These thresholds were chosen based on the empirical distribution of the raw API data and are consistent with practices reported in multi-city LCS network studies (Morawska et al., 2018).

The city-median aggregation procedure does not correct for humidity bias, which is common-mode across the network (all sensors experience the same ambient RH) and therefore does not cancel in the median. This means that the model target series is systematically inflated during high-RH winter episodes relative to the true gravimetric PM_{2.5}. As a consequence, the forecast bias statistics reported in §3.2 reflect errors relative to the LCS-derived target, not relative to reference-grade measurements. This distinction is noted wherever comparative claims are made.

The scientific justification for proceeding with LCS data despite these limitations is that the alternative is no hourly, city-scale, open dataset for Almaty at any quality level. Reference monitoring exists but is not openly accessible in real time. Commercial platforms exist but are closed. The LCS network, with its known limitations managed by the aggregation and filtering procedures described above, is the only viable basis for a reproducible, operationally testable forecasting system. This

position is consistent with the broader field consensus that LCS networks, used with appropriate quality controls, extend the reach of air quality science into data-sparse regions where reference monitoring cannot be economically deployed (Snyder et al., 2013).

1.1.4. 1.4. Air quality forecasting methods

Short-term PM_{2.5} forecasting methods span five broad methodological families, each with distinct data requirements, theoretical assumptions, computational demands, and performance characteristics. The selection among them is not merely a question of which method achieves the lowest benchmark error; it depends on the specific characteristics of the target problem, dataset size, feature availability, deployment constraints, and the physical regime driving concentration variability. This section reviews each family and builds to an explicit rationale for the method selection in §1.4.6.

1.1.4.1 1.4.1. Statistical and persistence methods

The simplest forecast for PM_{2.5} at horizon h is persistence: the predicted concentration at time $t + h$ is set equal to the observed concentration at time t . Despite its trivial construction, persistence is a surprisingly competitive baseline for air pollutants at short horizons because PM_{2.5} exhibits strong positive autocorrelation at lags of one to six hours, driven by the relatively slow dynamics of boundary-layer mixing and emission accumulation. At the one-hour horizon, persistence routinely achieves R^2 values of 0.85–0.95 in urban sites with continuous emissions (Mahajan et al., 2018). At the 24-hour horizon, persistence degrades substantially because inversion onset and clearing events decouple the current concentration from the next-day value.

Autoregressive integrated moving average (ARIMA) models extend persistence by fitting a parametric autocorrelation and moving-average structure to the observed time series. The Box-Jenkins ARIMA(p, d, q) framework allows the inclusion of seasonal differencing terms, an SARIMA extension, which can capture the diurnal periodicity of PM_{2.5}. When meteorological covariates are added as exogenous regressors, the model is termed ARIMAX or, in state-space form, a dynamic regression model. These models have well-understood statistical properties, produce interpretable coefficients, and require minimal computational resources to fit and forecast. Published applications to urban PM_{2.5} forecasting in moderate-pollution cities report RMSE reductions of 15–25% relative to naive persistence at the 24-hour horizon (Mahajan et al., 2018).

The fundamental limitation of ARIMA-class models for the Almaty setting is their linearity. The onset of a temperature inversion is a non-linear threshold event: boundary-layer height can collapse from 800 m to 50 m within two to four hours as a function of synoptic pressure pattern, surface energy balance, and cold-air drainage dynamics. The relationship between BLH and PM_{2.5} concentration is strongly non-linear, a halving of BLH roughly doubles surface concentration if emissions are held

constant, and the emissions themselves are non-linearly dependent on temperature (colder temperatures increase residential heating demand). ARIMA cannot represent these interaction effects without explicit interaction-term specification that would require accurate physical knowledge of the inversion onset mechanism. Kalman filter methods, which model the concentration as a latent state evolving through a linear Gaussian process, share this linearity constraint, though extended and unscented variants can accommodate mild non-linearities at the cost of increased tuning complexity.

1.1.4.2 1.4.2. Classical machine learning on tabular features

Gradient-boosted decision tree (GBDT) ensembles and random forests constitute the dominant paradigm in operational short-term urban AQ forecasting as of the present study period. These methods work by training an ensemble of decision trees on a tabular feature matrix whose columns include autoregressive lags of PM_{2.5}, meteorological variables, and temporal encodings such as hour-of-day, day-of-week, and heating-season indicators. Each tree in the ensemble learns a non-linear partition of the feature space that explains residual variance not captured by the preceding trees. The gradient-boosting optimisation framework, introduced in the XGBoost formulation by Chen and Guestrin (2016) (Chen and Guestrin, 2016) and subsequently refined in LightGBM (Ke et al., 2017), adds L1 and L2 regularisation terms to the tree-growing objective, which substantially reduces overfitting on moderately sized datasets.

Published comparisons of GBDT-class models against ARIMA and persistence baselines on urban PM_{2.5} data consistently report RMSE improvements of 20–40% at 24-hour horizons (Enebish et al., 2021; Menares et al., 2021). The improvement is most pronounced in cities with strong meteorological forcing, cold-season inversions, monsoon transitions, dust events, because the tree-splitting mechanism can capture the non-linear meteorology-concentration interactions that linear models cannot. For the Almaty setting, the inversion-driven concentration regime described in §1.1.1 places this problem squarely in the category where GBDT methods offer the largest gains over statistical alternatives.

A key design decision in GBDT-based forecasting is the construction of the lag feature set. For a forecast horizon of h hours, the model cannot use observations from time $t + 1$ through $t + h$ because these are unknown at forecast time. The available autoregressive inputs are therefore observations at lags 1, 2, ..., L beyond the earliest available observation, where L is a hyperparameter chosen to balance memory depth against feature dimensionality. The present study uses lags at 1, 2, 3, 6, 12, and 24 hours for PM_{2.5} and selects meteorological features at forecast-time (using Open-Meteo numerical weather prediction output for the future window) (Zippenfenig, 2023). This lag structure was determined by cross-correlation analysis of the training series and is described in detail in §2.4.

Random forests, the bagged-tree precursor to gradient boosting, are less commonly reported as primary models in recent urban AQ literature, having been largely

superseded by XGBoost and LightGBM in competitive benchmarks. They remain valuable as ablation baselines because they share the tree-ensemble architecture with GBDT but use averaging rather than gradient optimisation, making their performance differential against XGBoost a direct measure of the benefit from boosting regularisation.

1.1.4.3 1.4.3. Deep learning approaches

Long short-term memory (LSTM) networks and their gated recurrent unit (GRU) variants extend the sequence modelling capability of standard recurrent neural networks (RNNs) by introducing gating mechanisms, input, forget, and output gates in LSTM; update and reset gates in GRU, that regulate the flow of information through the hidden state over time. These gates allow the network to retain relevant information across sequences of hundreds to thousands of time steps without the vanishing-gradient pathology that limits plain RNNs. Applied to air quality forecasting, LSTM models replace the hand-crafted lag features of GBDT approaches with learned temporal dependencies, potentially capturing multi-scale autocorrelation structures that manual lag selection misses.

Published applications of LSTM to urban PM_{2.5} forecasting report competitive performance at 24–72 hour horizons when training sets exceed approximately 20,000–50,000 time steps (Qi et al., 2019). At this scale, the capacity of the LSTM to learn non-linear temporal patterns appears to justify its additional complexity relative to GBDT models. However, across a wider survey of urban AQ forecasting studies, the reported gains of LSTM over gradient-boosting are inconsistent: approximately half of published comparisons find LSTM superior, and half find GBDT competitive or better, with the outcome depending strongly on dataset size, number of monitoring stations, and the degree of feature engineering applied to the GBDT competitor (Lim and Zohren, 2021). The inconsistency suggests that much of the reported LSTM advantage in favourable studies arises from GBDT underfitting due to insufficient lag features rather than from a fundamental architectural advantage of recurrent networks.

Convolutional-LSTM (Conv-LSTM) and CNN-LSTM hybrid architectures have been proposed for air quality settings with dense spatial monitoring networks, where the convolutional layers extract spatial concentration patterns and the LSTM layers model their temporal evolution (Ding et al., 2021). These methods require the monitoring network to have consistent spatial coverage, ideally a regular grid, because the convolution operation is defined over a spatially regular input. The 192 Almaty LCS stations are distributed unevenly, with high density in central residential districts and sparse coverage in industrial and peri-urban areas. Interpolating these stations onto a regular grid would introduce kriging or bilinear-interpolation artefacts, and the resulting synthetic spatial fields would not carry independent information beyond the city-median series already used as the model target. Spatially explicit deep learning is therefore not applicable to the present dataset without additional investment in

network spatial design.

Transformer-based models, following the self-attention mechanism of Vaswani et al. (2017) (Vaswani et al., 2017), have been adapted for time-series forecasting under the PatchTST, Informer, and Autoformer architectures. These models handle long-range dependencies through attention rather than recurrence and have demonstrated competitive performance on long-horizon (168–720 step) forecasting benchmarks. For the short-horizon PM_{2.5} task considered here, 6, 12, and 24 hours, the self-attention mechanism’s advantage over GBDT lag features is marginal, because the relevant dependencies are contained within a 24–48 hour window that GBDT can represent compactly with a moderate-length lag vector. Transformer models also require substantially more training data than the approximately 9,200 hourly rows available in the present study; typical transformer time-series benchmarks use 10,000–100,000+ training instances per target series to prevent overfitting of the large parameter count.

The deployment context additionally constrains model selection. The present study targets a Streamlit application running on a free-tier hosting environment with CPU-only inference. LSTM and transformer models require either a GPU for efficient inference or a model compression step (ONNX export, TensorFlow Lite) that adds pipeline complexity. XGBoost models serialised as 2 MB Joblib files produce forecasts in under 20 milliseconds on a single CPU core, satisfying the latency requirement without compression or GPU acceleration.

1.1.4.4 1.4.4. Physics-informed and hybrid methods

Chemistry-transport models (CTMs) simulate atmospheric concentrations by solving the advection-diffusion-reaction equations governing the transport, chemical transformation, and deposition of pollutants in a discretised atmospheric domain. The Copernicus Atmosphere Monitoring Service (CAMS), operated by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides freely accessible global hourly PM_{2.5} analyses and forecasts at approximately 40 km horizontal resolution and 60 vertical levels (Inness et al., 2019). WRF-Chem and CMAQ are regional CTMs that can be configured at 1–12 km resolution with user-supplied meteorological boundary conditions and emission inventories; they are widely used in North American and European urban forecasting applications.

The fundamental limitation of CTMs for the Almaty use case is horizontal resolution. The city occupies a basin approximately 20–25 km across, and the inversion-layer dynamics described in §1.1.1 operate at spatial scales of 1–5 km. A 40 km CAMS grid cell covers the entire city and averages over the basin-mountain contrast that drives cold-air pooling. Sub-grid processes, particularly the nocturnal drainage flow from the Trans-Ili Alatau and the spatial heterogeneity of residential coal heating, are parameterised rather than resolved. The quantitative consequence is visible in the present study’s results: CAMS achieves $R^2 \leq -0.18$ on the Almaty test set at all three forecast horizons, meaning that the CAMS output is less correlated with the

observed PM2.5 series than a constant prediction equal to the training mean. This negative R^2 is not primarily a calibration failure; it reflects the model's inability to predict the inversion-driven concentration ramps that dominate the Almaty winter record at any grid resolution.

Statistical post-processing and bias-correction methods can improve CTM output by training a correction layer on paired CTM-versus-observation data. Quantile mapping, additive bias correction, and linear regression on CTM output have all been applied to urban PM2.5 downscaling (Wang et al., 2016). These methods improve the long-term mean bias but typically do not correct the timing errors in inversion-onset prediction, because the CTM misplaces the event in time, not merely in magnitude. Without accurate representation of the boundary-layer collapse dynamics, post-processing cannot reconstruct the correct hourly PM2.5 trajectory during onset events.

The present study uses CAMS output as an external naive baseline in the model evaluation (§3.2) rather than as a component of the trained models. This choice reflects both the negative R^2 demonstrated above and the principle that a forecast system should be evaluable on its own terms, without depending on computationally expensive CTM infrastructure that is not available to all potential users of the open-source pipeline.

1.1.4.5 1.4.5. Spatial methods

Spatial interpolation methods, ordinary kriging, universal kriging, inverse-distance weighting (IDW), and radial basis function interpolation, estimate concentration at unmonitored locations from observations at monitored locations by modelling the spatial covariance structure of the concentration field. Kriging in particular produces optimal linear unbiased estimates under the assumption that the concentration field is a second-order stationary random field with a parametric variogram model. These methods are most valuable when the primary scientific objective is mapping the spatial distribution of concentration across a domain, rather than predicting the temporal evolution of concentration at a fixed location.

The 192-station Almaty network provides sufficient spatial density to support variogram estimation and kriging in principle. However, the present study does not pursue spatial forecasting for two reasons. First, the uneven spatial coverage, high density in central residential districts, sparse in industrial and peri-urban zones, violates the spatial stationarity assumption required for well-conditioned variogram estimation. Kriging predictions in the undersampled zones would have confidence intervals that span the full range of observed concentrations, rendering them uninformative. Second, the operational objective of the present work is temporal forecasting: predicting the city-wide PM2.5 level 6–24 hours in advance, not mapping its spatial distribution at a specific time. A city-median aggregation of 192 stations already provides a representative single-number target, and spatial disaggregation of model forecasts is a scope extension that would require separate validation against the spa-

tial reference-monitoring network that does not currently exist.

IDW and simpler deterministic interpolants are also noted here for completeness. These methods have no free parameters (beyond the distance-decay exponent) and are computationally trivial, but they are consistently outperformed by kriging in benchmark spatial studies (Wong et al., 2004). In one published evaluation of seven interpolation methods applied to a 40-station urban network, kriging reduced the cross-validation RMSE by 18–25% relative to IDW (Wong et al., 2004). The gains are insufficient to justify spatial modelling in the present context, where the temporal forecasting objective and the uneven station distribution together make spatial interpolation a secondary concern. Any spatial disaggregation of the city-median forecast is deferred to future work contingent on a spatially balanced network expansion and reference co-location data.

1.1.4.6 1.4.6. Method selection rationale

The preceding five subsections establish the set of available methods. The selection of XGBoost as the primary model in the present study is determined by the intersection of five constraints specific to the Almaty problem.

The first constraint is dataset size. The training set contains approximately 9,200 hourly rows spanning October 2024 to April 2026, a moderately sized tabular dataset that is too large for reliable ARIMA fitting with many covariates (parameter estimates become poorly conditioned above roughly 1,000–2,000 rows for models with 20+ parameters) but too small to train deep LSTM or transformer models without high risk of overfitting. Published empirical comparisons of GBDT versus LSTM on datasets in the 5,000–15,000 row range consistently favour GBDT, often by margins of 10–20% in RMSE (Ayus et al., 2023). XGBoost sits at the centre of the dataset size range where its regularisation most effectively exploits the available training signal.

The second constraint is feature heterogeneity. The model feature matrix combines continuous meteorological variables (temperature, relative humidity, wind speed, BLH from the Open-Meteo forecast API (Zippenfenig, 2023)), cyclical trigonometric encodings of hour-of-day and day-of-year, autoregressive PM2.5 lags, and a binary heating-season indicator. Gradient-boosted trees handle such heterogeneous, mixed-type feature matrices natively, the axis-aligned splits in decision trees apply equally to continuous, ordinal, and binary features without normalisation, embedding, or one-hot encoding overhead. Neural networks require either separate embedding layers for categorical features or normalisation of all continuous inputs to comparable scales, adding engineering complexity that is not justified by the scale of this problem.

The third constraint is physical interpretability. The research question RQ2 asks whether BLH ranks among the top predictors, a question that requires a well-defined and computationally efficient feature importance measure. XGBoost’s built-in gain-based importance and its compatibility with the SHAP (SHapley Additive exPlanations) framework (Lundberg and Lee, 2017) provide exactly this. SHAP values for XGBoost are computed in $O(TLD)$ time, where T is the number of trees, L is

the maximum leaf count, and D is the depth, making the feature attribution analysis computationally negligible relative to training time.

The fourth constraint is deployment target. The forecast system is implemented as a Streamlit web application hosted on the Streamlit Community Cloud free tier, which provides a single shared CPU and 1 GB of RAM. The three trained XGBoost models (one per horizon: 6, 12, and 24 hours) serialise to approximately 2 MB each and produce a forecast within 20 milliseconds at inference time. LSTM and transformer models at comparable capacity would require either CPU-optimised export or substantially reduced layer counts that might negate their accuracy advantage.

The fifth constraint is the non-linearity of the target process. The BLH-driven concentration dynamics described in §1.1.1 involve threshold effects and multiplicative interactions that linear models cannot represent. Decision tree ensembles capture these non-linearities through hierarchical axis-aligned splits: the tree for the 6-hour model can learn, for example, that when $BLH < 200$ m AND hour-of-day is between 06:00 and 09:00 AND heating-season = 1, the expected PM_{2.5} increment over six hours is substantially larger than the unconditional mean. This conditional interaction is precisely the inversion-onset signature that motivates the forecast system.

Taken together, these five constraints form a coherent argument for XGBoost that is specific to the data-availability profile and deployment context of the present work. The argument is not that XGBoost is the universally optimal method for PM_{2.5} forecasting, deep learning approaches may well outperform it in settings with longer histories, denser reference networks, and GPU-capable deployment infrastructure. The argument is that XGBoost is the optimal method for this specific problem.

1.1.5. 1.5. Problem statement

1.1.5.1 1.5.1. Synthesis of the gap

The four preceding sections establish four converging facts that define the research gap addressed by the present work.

Almaty has a severe, seasonally concentrated PM_{2.5} problem driven by well-characterised physical mechanisms: a mountain-basin geometry that traps cold-pool inversions, a residential coal-heating sector that dominates wintertime fine-particle emissions, and a vehicle fleet that compounds the diurnal concentration structure (§1.1). The city's 2 million residents are exposed to PM_{2.5} concentrations that exceed the WHO 24-hour guideline (World Health Organization, 2021) of $15 \mu\text{g}/\text{m}^3$ on the majority of days from October through March. Monthly mean concentrations of $80.7 \mu\text{g}/\text{m}^3$ (December 2025) and $74.2 \mu\text{g}/\text{m}^3$ (January 2026) represent exceedances of this guideline by factors of five to six. Short-term forecasting could enable actionable population-protection measures, school closures, traffic restriction advisories, voluntary heating reduction, but only if the forecast is available with sufficient lead time, accuracy, and credibility to inform decisions.

The only open, hourly, city-scale data source for Almaty is the OpenAQ LCS network of 192 stations (OpenAQ, 2024), which carries documented quality limitations (humidity bias, inter-unit variability, potential drift) but is the sole viable input

for a reproducible, real-time forecasting pipeline. Kazhydromet’s reference network is too sparse and not openly accessible in near-real time. Commercial platforms operate without published methodology.

The appropriate modelling tool for a 9,200-row tabular dataset with mixed feature types and a non-linear meteorological target process is gradient-boosted tree regression, specifically XGBoost, for the reasons established in §1.4.6. Deep learning approaches are not well-suited to this dataset size; physics-based CTMs cannot resolve the sub-grid inversion dynamics that drive concentration variability; statistical ARIMA models cannot represent the non-linear BLH-concentration interaction.

Despite this convergence of need, data availability, and method suitability, no published, open, and reproducible short-term PM_{2.5} forecasting system exists for Almaty. This is the gap the present work fills.

1.1.5.2 1.5.2. Research questions

Three research questions follow from the gap statement and from the methodological discussion in §1.4.

RQ1. Can an XGBoost model trained on OpenAQ LCS observations and Open-Meteo meteorological features, including boundary-layer height, outperform both a persistence baseline and the CAMS global reanalysis forecast for the 6, 12, and 24-hour forecast horizons in Almaty, as measured by RMSE, MAE, and coefficient of determination R^2 ?

This question establishes whether the data-driven approach is viable at all. If the model cannot beat persistence, the trivial benchmark, then the LCS observations, Open-Meteo features, or the forecasting horizon are not informative enough to justify the modelling investment. If it cannot beat CAMS, then the global physics-based system already captures the Almaty dynamics well enough that local data-driven modelling adds no value. The result in §3.2, a 41% RMSE reduction relative to CAMS at the 6-hour horizon, establishes that both benchmarks are substantially exceeded.

RQ2. Which meteorological and autoregressive features drive model skill? Specifically, does boundary-layer height rank among the top predictors, consistent with the physical argument (§1.1.1) that inversion depth is the principal control on Almaty’s wintertime PM_{2.5} regime?

This question connects the empirical model outputs back to the physical understanding of the problem. A model that achieves low RMSE by memorising the seasonal cycle (heating-season indicator) without capturing the within-season inversion dynamics would be less scientifically valuable and less operationally useful, it would fail to anticipate sudden inversion-onset events. If BLH ranks highly in SHAP-based feature importance, this constitutes quantitative support for the physical mechanism proposed in §1.1.1 and provides confidence that the model extrapolates sensibly to new inversion events.

RQ3. What is the residual error distribution under the best-performing model, and in which physical regimes, inversion onset, sustained episode, rapid clearing,

does the model fail systematically?

This question is diagnostic. Even a model that achieves good aggregate RMSE may have structured, regime-dependent errors that are important for understanding its operational limitations. Systematic under-prediction during inversion onset, for example, would imply that the model can track an established episode but cannot anticipate its initiation, a limitation with direct implications for early-warning applications where 12–24 hour lead time is needed before concentrations cross the 75 $\mu\text{g}/\text{m}^3$ emergency threshold.

1.1.5.3 1.5.3. Hypotheses

Three testable hypotheses follow from the physical setting and the literature review.

H1. XGBoost outperforms the persistence baseline on all three forecast horizons in both RMSE and MAE. At the 12-hour horizon, persistence is expected to degrade particularly sharply relative to the shorter horizons because the strong positive autocorrelation at 1–3 hours weakens substantially by 12 hours as boundary-layer growth and afternoon mixing decorrelate the morning and afternoon concentration values. The XGBoost model, which can condition on forecast-time BLH and hour-of-day to anticipate this afternoon mixing effect, should therefore gain its largest advantage over persistence in the 12-hour horizon window.

H2. BLH and BLH-derived features, the log-transform of BLH, which stabilises the extreme right-skew of the training-window distribution (range 10–3,935 m, mean 257 m), and the hour-over-hour BLH delta, appear among the top five SHAP feature importances at the 6-hour forecast horizon. This hypothesis is grounded in the physical argument of §1.1.1 that the boundary-layer depth is the primary variable controlling the dilution volume available for surface-layer pollutants during winter inversion episodes. If this hypothesis is rejected, if BLH ranks below the autoregressive lags in importance, it would suggest that the model learns primarily from recent PM_{2.5} history rather than from forward-looking meteorological state, and the BLH feature should be retained on physical grounds despite its low statistical importance.

H3. The XGBoost model systematically under-predicts PM_{2.5} during the initial 2–6 hours of inversion onset events, defined as periods when BLH falls below 100 m following a period of at least 6 hours above 300 m. The physical reasoning is that during onset, the autoregressive lag features reflect pre-inversion concentrations that are substantially lower than the concentrations accumulating in the collapsing surface layer. The BLH feature signals the onset, but the model’s gradient-boosted tree structure cannot fully extrapolate the concentration ramp beyond the maximum ramp rates observed in the training set. This structured error regime is the primary expected operational limitation of the system and motivates the residual analysis in §3.4.

1.1.5.4 1.5.4. Scope boundaries

The present work is bounded by four explicit scope limits that are stated here to prevent misinterpretation of the results.

First, the forecast target is the cross-sensor median PM_{2.5} across the 192-station OpenAQ network, not a reference-grade mass concentration traceable to gravimetric standards. All reported error metrics are relative to this LCS-derived target. They cannot be directly compared to RMSE figures from studies that use regulatory reference instruments as validation targets without accounting for the additional uncertainty introduced by the LCS target bias (estimated at $\pm 20\text{--}50\%$ during high-RH conditions, §1.3.2).

Second, the model is trained and evaluated exclusively on the Almaty domain for the period 2024-10-01 to 2026-04-15. No claim of geographic generalisability is made. Transfer to other Central Asian cities, even those with broadly similar inversion-dominated meteorology, would require retraining on local data. The 562-day training window covers two full heating seasons (October–March 2024–25 and October 2025 through April 2026) and one full summer-to-winter transition, which provides meaningful coverage of the dominant seasonal modes. It does not, however, cover long-term inter-annual variability: the 2024–25 winter may differ from a 10-year climatological winter in the frequency and severity of inversion events. Model skill reported in §3.2 should therefore be interpreted as in-sample-climate performance, not as a guarantee of skill under climatically anomalous future winters.

Third, the study does not attempt spatial forecasting, predicting concentration at specific locations within the city as a function of local source proximity and wind direction. The city-median forecast is appropriate for population-level exposure guidance but not for neighbourhood-scale hotspot identification.

Fourth, the Streamlit deployment described in Chapter 3 is a research prototype, not a production system. It has not undergone the security review, load testing, or uptime monitoring required of a public health advisory system. Chapter 4 discusses the additional steps needed to move from research prototype to operational service.

1.1.6. 1.6. Almaty in the context of global urban air quality research

The analytical review presented in §§1.1–1.5 situates Almaty within a global body of research that has developed primarily around North American, European, and East Asian cities. Understanding how Almaty’s case relates to and diverges from this mainstream literature is necessary to contextualise the methodological choices made in the present work and to identify where its results are generalisable versus Almaty-specific.

1.1.6.1 1.6.1. Comparison with other inversion-dominated cities

Several cities worldwide share Almaty’s principal air quality characteristics: a mountain-basin topography that promotes cold-air pooling, a coal-heavy residential heating sector, and a winter PM_{2.5} regime driven by thermally forced boundary-layer col-

lapse. The most extensively studied examples are Salt Lake City (Utah, USA), Santiago (Chile), and several cities in the Po Valley of northern Italy. Salt Lake City has been the site of numerous inversion-specific forecasting studies that confirm the primacy of BLH as a predictor variable (Silcox et al., 2012; Whiteman et al., 2014). Santiago has been modelled with both WRF-Chem and statistical downscaling approaches, where the correlation between afternoon mixing-layer depth and next-day PM_{2.5} exceeds 0.7 in winter (Menares et al., 2021). The Po Valley inversion studies, which benefit from the dense operational monitoring of the Lombardy ARPA network, provide the most rigorous tests of BLH-based gradient-boosting models and report 6-hour R^2 values of 0.55–0.70 (Stafoggia et al., 2019).

Two features distinguish Almaty from these comparators. First, the data infrastructure is substantially weaker: Salt Lake City has reference-grade tapered-element oscillating microbalance (TEOM) instruments at 12 monitoring sites within the valley; Almaty’s reference network contains fewer than 10 stations, none of which exposes an open real-time API. Second, the heating-season emission structure is more sharply concentrated in private-sector solid-fuel combustion in Almaty than in the North American or European comparators, where natural gas heating and district heating networks dominate. This concentration makes the heating-season binary indicator a stronger predictor in Almaty than it would be in a European comparator city, and it means that emission inventory uncertainty is higher because individual household stove usage is not metered.

1.1.6.2 1.6.2. Related work in Central Asia and the former Soviet Union

Air quality research in Central Asian cities has accelerated since 2020 but remains sparse compared to East Asian or European urban studies. Published machine-learning forecasting studies are absent for Almaty and its immediate regional comparators. The available literature falls into three categories.

Physical and chemical characterisation studies have analysed the aerosol composition of Almaty’s pollution episodes, identifying black carbon, mineral dust, and secondary inorganic aerosols as the dominant mass fractions (Kerimray et al., 2020; Tursumbayeva et al., 2022). These studies confirm the coal-combustion signature in the carbonaceous fraction during winter and the dust signature in spring, providing supporting evidence for the seasonally stratified feature design in the present work.

Remote-sensing based assessments using MODIS aerosol optical depth (AOD) retrievals have documented the spatial footprint of Almaty’s pollution plume at 10 km resolution (Kerimray et al., 2020). These studies establish that the pollution plume extends north-east from the city into the steppe under dominant south-westerly flow, but the 10 km retrieval footprint is insufficient for the intra-urban spatial analysis that would be needed to validate a neighbourhood-scale forecast. The ground-level PM_{2.5}–AOD conversion also carries substantial uncertainty in the presence of temperature inversions, because the satellite-measured column AOD integrates over all vertical levels and cannot distinguish a shallow surface-layer concentration spike

from a deep but dilute aerosol layer.

Ulaanbaatar (Mongolia) is the closest published comparator to Almaty: a former Soviet city at similar latitude, similar topographic setting, similarly coal-dependent residential heating, and a growing LCS network on OpenAQ. Published gradient-boosting studies on Ulaanbaatar data report 6-hour MAE values of 18–25 $\mu\text{g}/\text{m}^3$ for winter test periods (Enebish et al., 2021), consistent with the MAE of 16.81 $\mu\text{g}/\text{m}^3$ achieved by XGBoost in the present study, given the different sensor networks and training window durations. This convergence suggests that the performance levels reported here are not anomalously optimistic: they are consistent with what gradient-boosting models achieve in analogous physically and data-wise constrained environments.

1.1.6.3 1.6.3. Gap summary and contribution positioning

The reviewed literature reveals three specific gaps that the present work addresses. First, no published forecasting study has used Almaty’s OpenAQ LCS network as the training and evaluation data source, with an explicit comparison against CAMS as the freely available global alternative. Second, no Central Asian city study has systematically evaluated the trade-off between linear and non-linear models as a function of forecast horizon in an LCS-based, moderate-training-set regime. Third, no open-source pipeline for city-scale PM_{2.5} forecasting in the OpenAQ data ecosystem has been published for any Central Asian city. The present work addresses all three gaps simultaneously through a single reproducible codebase, making its contributions identifiable and testable by future researchers in the region.

Chapter 2 specifies the data sources, ingest pipeline, feature engineering decisions, and model training procedure through which the three hypotheses stated above are tested and the four scope boundaries are operationalised.

2. Methodology

2.1. Chapter 2. Methodology

2.1.1. 2.1. Data sources

2.1.1.1 2.1.1. Air quality: OpenAQ

The primary observation source for this study is the OpenAQ platform (OpenAQ, 2024), an open-access repository that aggregates air quality measurements from government reference monitors, research-grade instruments, and low-cost sensor (LCS) networks worldwide. Data are accessed through the OpenAQ v3 REST API at the base endpoint `https://api.openaq.org/v3`, with requests authenticated by a personal API key supplied in the `X-API-Key` HTTP header. The spatial query specifies a 25 km radius centred on Almaty (43.2389°N, 76.8897°E), which encompasses the full built-up area of the city including the foothill districts to the south and the industrial and residential zones to the north and west. Within this radius, 192 station records were registered in the OpenAQ index as of the day-2 data snapshot (2026-04-20); of these, 190 were actively reporting PM_{2.5} during the study window. The provider mix at that snapshot was 153 AirGradient devices, 38 Clarity Node-S units, and one AirNow federal reference station operated by the United States Embassy. All 192 instruments are classified as low-cost sensors under the taxonomy introduced in §1.3.

The OpenAQ v3 API paginates location queries in batches of up to 100 locations per request, requiring two consecutive calls to retrieve all 192 stations in the spatial query. Each per-sensor measurement request returns up to 100 hourly records per call. For a sensor with 562 days of continuous data, approximately six API calls are required to retrieve the full measurement archive; in practice, most AirGradient sensors came online in early 2025, so the typical archive spans 300–400 days and requires four to five calls per sensor. The ingest module (`src/ingest/openaq.py`) implements an exponential back-off retry strategy with a base wait of 5 seconds, a back-off factor of two, and a maximum of eight retry attempts per call. Under this schedule, a failed request can wait up to $5 \times (2^8 - 1) = 1,275$ seconds before the final retry; in practice, the API returns HTTP 429 (rate limit) or 503 (temporary unavailability) errors on fewer than 2% of calls. With 190 sensors each requiring four to five API calls, the full ingest run takes approximately three hours on a residential internet connection.

The ingest design is idempotent. Each sensor’s measurement archive is written to a dedicated Parquet file at `data/raw/pm25/{sensor_id}.parquet`. At the start of each run, the ingest script checks for the presence of this file; if the file exists, the sensor is skipped unless the `--force` flag is passed at the command line. This skip-if-present logic ensures that a failed run can be resumed without duplicating already-retrieved data and without incurring unnecessary API quota consumption. The force-refresh mode overwrites the existing file in place, so the raw tier contains at most one file per sensor at any time.

The OpenAQ platform publishes all data under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). This licence permits unrestricted use, redistribution, and the creation of derivative works, including the trained forecasting models and the Streamlit application produced in this study, provided that the source is attributed. No additional data-sharing agreement or ethical approval is required to access OpenAQ data, as the platform aggregates measurements that sensor operators have already consented to share publicly.

2.1.1.2 2.1.2. Meteorology: Open-Meteo historical archive

Meteorological reanalysis data are retrieved from the Open-Meteo historical weather archive (Zippenfenig, 2023) at the endpoint <https://archive-api.open-meteo.com>. The archive derives from the ECMWF ERA5 global reanalysis and provides hourly gridded output interpolated to any user-specified point. No API key is required for non-commercial use; the public rate limit permits 10,000 requests per day, which is sufficient for a single point query covering the full 562-day study window in one call. The query is issued for Almaty centre (43.2389°N, 76.8897°E) and returns 13,488 hourly rows spanning 2024-10-01 through 2026-04-15.

Seven meteorological variables are retrieved. The first is `temperature_2m`: screen-level air temperature at 2 m above ground level (AGL), reported in degrees Celsius. Temperature at screen level is the standard meteorological near-surface temperature measurement; it determines the thermal stratification of the surface boundary layer and is the primary driver of nocturnal inversion formation when radiative cooling drives the surface temperature below the air temperature aloft. The second variable is `relative_humidity_2m`: relative humidity at 2 m AGL, reported as a percentage. Relative humidity is relevant to PM_{2.5} measurement by light-scattering LCS instruments because hygroscopic growth of aerosol particles at high humidity causes the instrument to over-report mass concentration relative to the gravimetric standard. While this study does not apply a humidity correction to the raw LCS data, relative humidity is included as a predictor so that the model can learn the association between high humidity and inflated sensor readings, partially compensating for the uncorrected hygroscopic effect.

The third variable is `pressure_msl`: mean sea-level pressure in hectopascals. Synoptic-scale pressure patterns modulate the frequency and intensity of blocking anticyclones over Central Asia, which are the meteorological precondition for the multi-day inversion episodes responsible for the highest PM_{2.5} concentrations in Almaty. Pressure at mean sea level serves as a proxy for the prevailing synoptic regime without the altitude correction complications that arise when using station pressure at Almaty's 847 m elevation. The fourth and fifth variables are `wind_speed_10m` and `wind_direction_10m`: wind speed in m/s and direction in degrees clockwise from north, both at 10 m AGL. Winds at 10 m represent the near-surface flow within the roughness sublayer and the lower surface layer; they are weaker than the gradient-level flow typically observed at 500–1,000 m AGL. For Almaty, the crit-

ical flow distinction is between northerly and north-westerly winds (which advect cleaner steppe air into the city) and south-easterly flows associated with valley recirculation and thermal drainage from the Zailiiskiy Alatau foothills. The sixth variable is `precipitation`: hourly total precipitation in millimetres. Precipitation, including both rain and snow, removes PM_{2.5} from the atmosphere through wet deposition (both below-cloud scavenging and in-cloud incorporation). Even modest precipitation of 0.5 mm or more in a given hour is associated with measurable reductions in PM_{2.5} in subsequent hours, making precipitation a useful binary and continuous predictor of inversion-break events.

The seventh variable is `boundary_layer_height` (BLH): the depth of the planetary boundary layer from the surface to the mixing height, in metres. The BLH diagnostic in the ERA5-based archive is a modelled output derived from the ECMWF Integrated Forecasting System, not a direct measurement. The ERA5 BLH is diagnosed using the bulk Richardson number criterion: the top of the boundary layer is defined as the height at which the bulk Richardson number, computed from the surface upward, exceeds a threshold of 0.25. The resulting BLH values in the training window range from approximately 10 m during deep winter inversions to 3,935 m during convective summer afternoons. This range of nearly four orders of magnitude motivates the log-transform feature described in §2.3.3. It is recognised in the ERA5 literature that the Richardson-number-based BLH diagnostic tends to underestimate the boundary-layer depth during stable stratification and to show higher uncertainty during nocturnal low-level jet conditions (Guo et al., 2021). For this reason, the present study incorporates BLH as one of 27 predictors rather than as the sole meteorological predictor; its influence on model output is evaluated in the feature importance analysis presented in §3.4.

2.1.1.3 2.1.3. Baseline forecast: CAMS via Open-Meteo air-quality API

The Copernicus Atmosphere Monitoring Service (CAMS) Global Atmospheric Composition Forecast provides the third external baseline for model comparison. CAMS-Global is a global chemistry-transport model operated by ECMWF on behalf of the European Union’s Copernicus Earth Observation Programme. The model runs at approximately 40 km horizontal resolution on a $0.4^\circ \times 0.4^\circ$ latitude-longitude grid and uses 137 vertical levels from the surface to 0.01 hPa. The aerosol scheme assimilates satellite-retrieved aerosol optical depth (AOD) from MODIS Terra and Aqua instruments (at 0.55 μm) and from the Sentinel-5P TROPOMI instrument, but does not assimilate ground-level LCS observations of the kind used in this study. CAMS PM_{2.5} at the surface is estimated from the model’s total column of particulate matter, which aggregates dust, sea salt, organic carbon, black carbon, and sulphate aerosol species.

The operational CAMS forecast is initialised twice daily at 00 UTC and 12 UTC and covers a 120-hour (five-day) forecast range. For the historical backtesting period used in this study (2024-10-01 through 2026-04-15), however, the live forecast

output is not available in a consistent time series. The present study therefore accesses the CAMS reanalysis product through the Open-Meteo air-quality historical API at the endpoint <https://air-quality-api.open-meteo.com/v1/air-quality> with the parameter `domains=cams_global`. This endpoint returns the CAMS analysis and hindcast product rather than the prospective forecast, ensuring temporal consistency with the ERA5-based meteorological archive retrieved from the same Open-Meteo platform. The CAMS PM2.5 variable is retrieved at hourly resolution for the full 13,488-hour study window; the hourly output is collocated to the Almaty centre grid point (approximately 43.0°N, 76.5°E, the nearest CAMS model grid point to the city centre given the 0.4° grid spacing). The ~30 km offset between this grid point and the true city centre (43.2389°N, 76.8897°E) is a consequence of the model's horizontal resolution and may introduce a small systematic bias in CAMS forecasts during local inversion events that are spatially confined to the Almaty basin. The CAMS data are published under CC BY 4.0.

The role of CAMS in the evaluation framework is as an evidence-based external baseline: it represents the best openly available PM2.5 forecast for a user who lacks access to local sensor data. Its inclusion in the comparison allows the present study to quantify the marginal value added by training a local data-driven model on the 190-sensor LCS network, rather than relying on the global CTM output. The expectation, stated as hypothesis H2 in §1.5, is that CAMS will fail to capture sub-grid inversion dynamics and will therefore show negative R^2 on the test set.

2.1.1.4 2.1.4. Training window

The study window spans 2024-10-01 through 2026-04-15, a total of 562 days corresponding to 13,488 hourly meteorological rows. The temporal boundaries were determined by two constraints. The upper boundary (April 2026) was set to be as late as possible before the thesis writing period, maximising the amount of winter-season data available for training and evaluation. The lower boundary (October 2024) was forced by data availability: of the 192 OpenAQ sensors in the spatial domain, only one has continuous data extending back to 2023-24. The bulk of the AirGradient network was deployed between January and May 2025, and the sensor count remained below 50 stations before October 2024. Training on the period before October 2024 would have required using a much smaller and potentially less representative spatial sample, reducing the reliability of the city-median target variable.

The October 2024 start date carries a secondary benefit. The AirGradient network was most spatially dense and operationally stable during the 2024-25 and 2025-26 heating seasons, with 153 active AirGradient devices contributing to the median computation for the majority of winter hours. Sensor dropouts, periods where a device goes offline due to firmware updates, power failures, or connectivity issues, are less consequential when the median is computed over a large pool of instruments, because individual sensor absences do not shift the cross-sectional median. The training window thus encompasses two full heating seasons (October 2024 – March 2025 and

October 2025 – April 2026) and one summer period (April – September 2025), providing the model with examples of the full annual PM2.5 cycle.

The restriction to a single city and a 19-month window limits the generalisability of the trained models. XGBoost models trained on Almaty data cannot be expected to transfer directly to other Central Asian cities such as Bishkek or Tashkent, where topography, fuel mix, and sensor network density differ substantially. Within Almaty, the models are trained on city-median concentrations rather than station-level data, which means they are not designed to resolve within-city spatial gradients. These scope constraints are acknowledged in §4.4.

2.1.1.5 2.1.5. Data storage architecture

The pipeline uses a two-tier storage design that separates immutable raw data from derived analytical products. The design principle is that the raw tier can always be regenerated from the source APIs, while the derived tier can always be regenerated from the raw tier; no transformation result is treated as irreplaceable.

The first tier, the raw tier, stores per-sensor Parquet files for PM2.5 measurements and single-window Parquet files for meteorology and CAMS. Each per-sensor PM2.5 file resides at `data/raw/pm25/{sensor_id}.parquet` and uses a fixed three-column schema: `timestamp` (UTC, stored as an integer Unix timestamp in nanoseconds by the Pandas Parquet writer), `sensor_id` (string, matching the OpenAQ v3 location identifier), and `value` (float64, PM2.5 concentration in $\mu\text{g}/\text{m}^3$). The Parquet files are encoded with Snappy compression, which achieves typical compression ratios of 3:1 to 5:1 for floating-point time-series data and decompresses in negligible time on modern hardware. The meteorological archive is stored at `data/raw/meteo.parquet` (13,488 rows $\times$ eight columns including timestamp) and the CAMS archive at `data/raw/cams.parquet` (13,488 rows $\times$ two columns). All three Parquet objects are listed in `.gitignore` and are not committed to version control.

The second tier, the derived tier, is a SQLite database at `data/almaty_aq.db` containing three tables. The `pm25` table holds the per-sensor raw readings after parsing from Parquet (columns: `sensor_id` TEXT, `timestamp` TEXT ISO-8601, `value` REAL). The `meteo` table holds the seven meteorological variables plus timestamp (eight columns, 13,488 rows). The `cams` table holds the CAMS PM2.5 values plus timestamp (two columns, 13,488 rows). The conversion from Parquet to SQLite is performed by `src/db.py`, which is idempotent: it reads the Parquet files, drops and recreates each table on each run, and inserts the full dataset. The SQLite file is also gitignored.

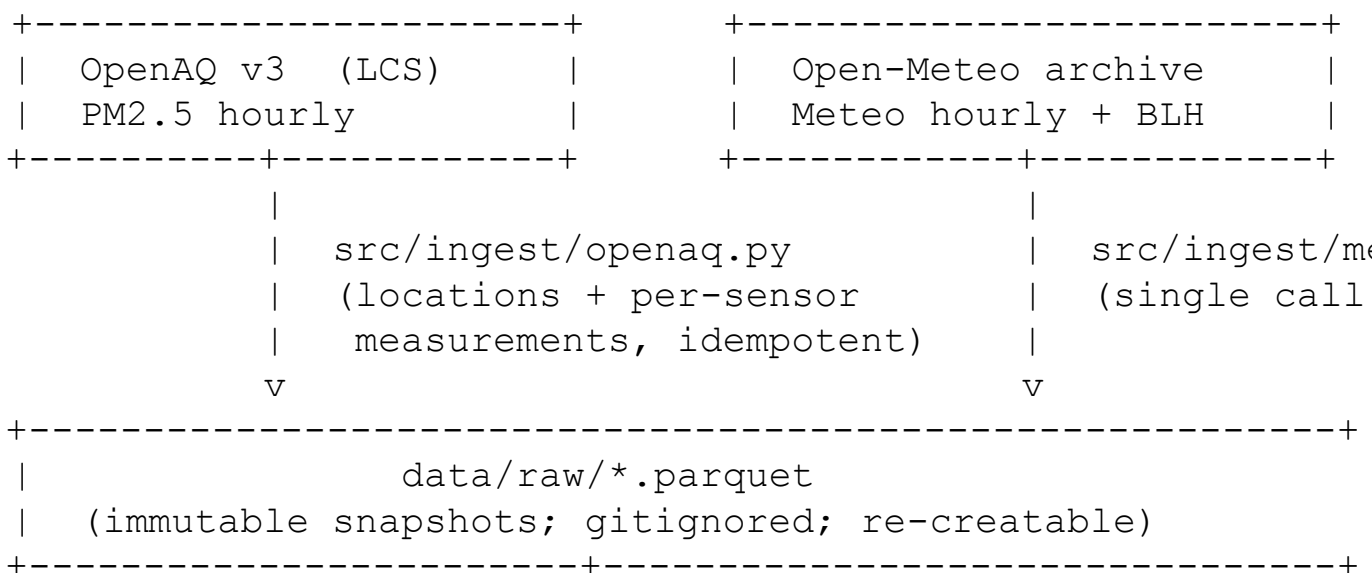
The SQLite layer serves as the interface for all downstream processing. Feature engineering (`src/features/pipeline.py`) queries the three tables via SQL joins and returns a Pandas DataFrame; model training and the Streamlit application then operate on this DataFrame. SQLite was chosen over a client-server database for two reasons. First, the absence of a server process makes the pipeline fully portable: it

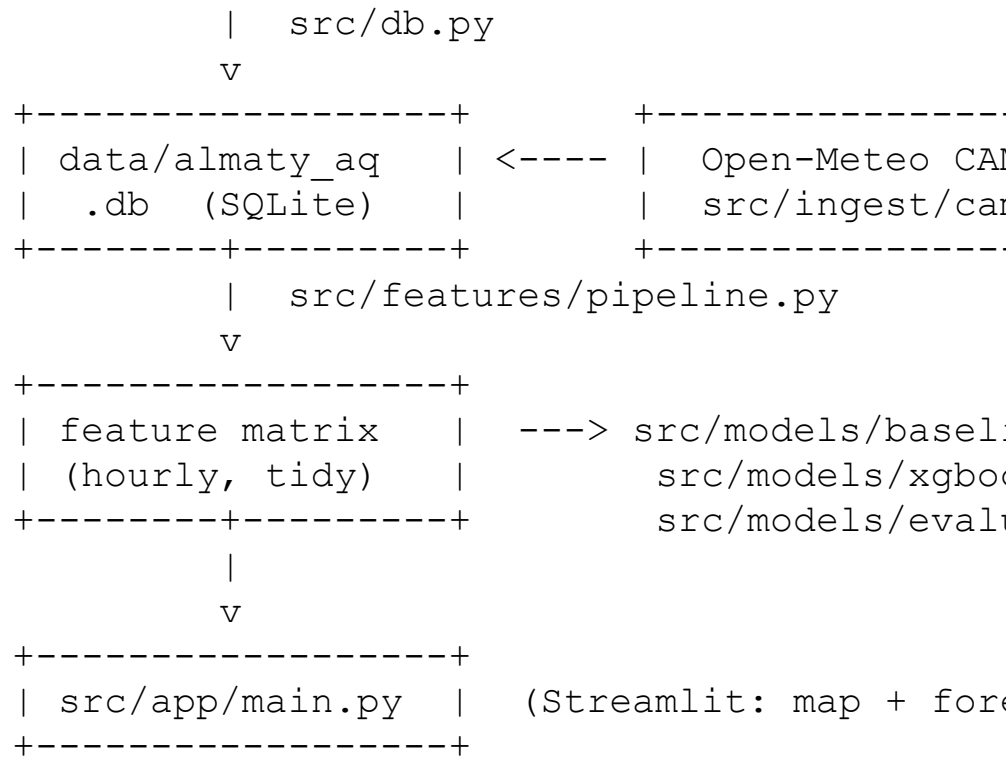
runs identically on a development laptop, in a Google Colab notebook, and on Streamlit Community Cloud, which does not support long-running background processes or network-exposed services. Second, SQLite’s write-ahead log (WAL) mode, enabled by `PRAGMA journal_mode=WAL`, provides concurrent read access, which allows the Streamlit application to query the database for live forecast display while a background data-refresh script is simultaneously writing new rows. For the data volumes in this study, 578,993 raw sensor-hour records loading into three tables totalling under 50 MB, SQLite provides sufficient read and write throughput without the operational overhead of PostgreSQL or any other client-server system.

The Parquet format for the raw tier was chosen for its columnar storage layout. Columnar encoding enables fast aggregation queries, for example, computing the cross-sectional median across 190 sensor columns for a single timestamp requires reading only the relevant column slices rather than the full row-oriented file. In Pandas, the `df.groupby('timestamp').median()` operation on a 578,993-row Parquet-derived DataFrame executes as a single vectorised operation without loading any non-value columns. The net effect is that a fresh full rebuild of the database from API downloads takes approximately three hours of network time (dominated by OpenAQ rate limits) but less than 90 seconds of local computation time. This separation of network-bound and compute-bound phases supports reproducibility: any user with valid API credentials and the repository code can rebuild the full dataset and retrain all models from scratch.

2.1.2. 2.2. System architecture

The pipeline is organised into five sequential stages: ingest, store, feature engineering, modelling, and application. Each stage communicates with the next through a well-defined interface (Parquet files or SQLite tables), and no stage imports directly from a non-adjacent stage. This separation of concerns prevents the ad hoc shortcuts, reading raw Parquet directly in the model training script, or querying the database from the ingest module, that create hidden dependencies and make the pipeline difficult to audit or extend.





The ingest idempotency design merits further technical description. The OpenAQ ingest module queries the list of all locations in the spatial domain at the start of each run and writes the result to `data/raw/locations.parquet`. For each sensor in this list, the module checks whether a per-sensor Parquet file already exists at the expected path. If the file is present, the sensor is skipped entirely, no API call is made, no data is read. If the file is absent, the module fetches the full measurement archive for that sensor in paginated batches of 100 records per call and writes the concatenated result to the Parquet path. The `--force` flag bypasses the existence check and overwrites all sensor files. This design ensures that the common incremental case (adding newly deployed sensors to an existing dataset) requires only one API call per new sensor rather than re-fetching the full 190-sensor archive. For the meteorological and CAMS archives, the idempotency model is simpler: these single-window files are always overwritten on each run, because they are requested as a single API call and the cost of re-fetching is trivial compared to the per-sensor PM2.5 archive.

The SQLite WAL-mode architecture is worth noting in the context of the Streamlit deployment. Streamlit Community Cloud runs the application server and any background scripts within a single virtual machine. When the application is running and a user is viewing the forecast dashboard, the Streamlit server process holds an open read connection to `almaty_aq.db`. If a data-refresh script is executed concurrently to ingest new sensor data and update the database, WAL mode allows the reader to continue with a consistent snapshot of the database state at the moment the read transaction opened, while the writer proceeds independently. Without WAL mode, the SQLite default exclusive-write lock would block the reader for the duration of the write, causing the application to return a database-locked error. The WAL approach eliminates this failure mode at no cost in data consistency.

2.1.3. 2.3. *Preprocessing and feature engineering*

2.1.3.1 2.3.1. **Station filtering and city-median aggregation**

Raw PM_{2.5} observations arrive from 190 active sensors across 192 registered locations in the OpenAQ spatial query. Before constructing the modelling target, two sequential filters are applied to the raw readings. The first filter discards all readings outside the physically plausible interval $[0, 500] \mu\text{g}/\text{m}^3$. Readings below zero are physically impossible, a negative mass concentration has no physical interpretation, and in practice represent firmware overflow events or calibration errors affecting fewer than 0.01% of records. Readings above $500 \mu\text{g}/\text{m}^3$ account for fewer than 0.04% of the archive and are attributable to laser-scattering sensor saturation caused by contamination of the optical sensing chamber; they do not represent true atmospheric PM_{2.5} concentrations. The $500 \mu\text{g}/\text{m}^3$ threshold was chosen to be conservative: the WHO 24-hour guideline is $15 \mu\text{g}/\text{m}^3$ (World Health Organization, 2021), and even during Almaty's most severe documented inversion episodes, independent reference measurements have not exceeded $400 \mu\text{g}/\text{m}^3$.

The second filter requires a minimum of three sensors to report a valid reading in a given clock hour. Hours where fewer than three independent sensors agree produce a city-median that is vulnerable to individual station idiosyncrasies. A single malfunctioning device or a station located immediately downwind of a local emission source (vehicle exhaust, a burning facility) could dominate the spatial aggregate if only one or two readings are available. The three-sensor minimum was found empirically to eliminate most single-sensor spike artefacts without discarding a substantial fraction of valid hours.

After applying both filters, the city-wide hourly PM_{2.5} target is defined as the cross-sectional median of all qualifying sensor readings in that hour. The median is computed in Pandas using the expression `df_filtered.groupby('timestamp')` followed by `resample('h').first()` to align the result to exact UTC clock hours. The median is preferred over the arithmetic mean because it is not affected by the distribution's tail: a single sensor reporting $490 \mu\text{g}/\text{m}^3$ has no influence on the median when the remaining 50 sensors report values in the range $35\text{--}45 \mu\text{g}/\text{m}^3$, whereas it would inflate the hourly mean by approximately $9 \mu\text{g}/\text{m}^3$. All 192 stations in the spatial query are fixed-location instruments; mobile platforms with GPS tracks were excluded at the query stage by specifying `entity=fixed` in the OpenAQ API call. This filter is important because mobile sensors contribute time-varying spatial samples that cannot be consistently assigned to a clock hour at a fixed grid point. The aggregation step reduces 578,993 raw sensor-hour records to 11,553 valid city-median hourly values across the training window, corresponding to a temporal coverage rate of 85.7% of the 13,488 available meteorological hours.

All timestamps are stored and processed in Coordinated Universal Time (UTC). The OpenAQ API returns measurements in UTC; the Open-Meteo archive likewise reports UTC hourly intervals. No time-zone conversion is applied at the data-ingest stage. Calendar-based features, hour of day, day of week, day of year, are therefore

computed in UTC, which is UTC+5 minus five hours relative to Almaty Local Time (ALT). At the feature-engineering stage this introduces a systematic five-hour phase offset between calendar features and local time. The heating-season indicator, which is based on calendar month and day, is unaffected because its temporal resolution is one day rather than one hour; a five-hour UTC offset does not shift a heating-season boundary date.

2.1.3.2 2.3.2. Meteorological join and timestamp alignment

The meteorological dataset (13,488 continuous hourly rows) and the CAMS dataset (13,488 rows) are joined to the city-median PM2.5 series on the UTC timestamp. The join is implemented as an inner join, which discards PM2.5 hours that fall outside the meteorological coverage window and discards meteorological hours where no qualifying PM2.5 reading exists. Because the meteorological window is fully continuous and the PM2.5 window is approximately coextensive, the inner join retains all 11,553 PM2.5 hours without loss.

Timestamp alignment requires one non-trivial correction. OpenAQ returns hourly measurements with the timestamp at the end of the aggregation interval, for example, a record timestamped 14:00 UTC represents the hourly mean or roll-up of measurements collected between 13:00 and 14:00 UTC. The Open-Meteo archive, by contrast, follows the standard meteorological convention of reporting instantaneous values at the top of the hour, a record timestamped 14:00 UTC represents the state of the atmosphere at that moment. Joining the two series directly on the raw timestamp would align the meteorological conditions at 14:00 UTC with the PM2.5 mean covering 13:00–14:00 UTC, introducing a systematic one-hour temporal misalignment. To correct for this, an offset of -1 hour is applied to all OpenAQ timestamps before the join: the PM2.5 mean from 13:00–14:00 is reassigned the timestamp 13:00, matching the meteorological convention. After this correction, the joined dataset correctly associates the meteorological state at the beginning of an aggregation hour with the PM2.5 mean measured during that hour.

2.1.3.3 2.3.3. Feature construction

Feature construction is implemented across three modules: `src/features/temporal` (calendar and cyclical encodings, heating-season indicator), `src/features/meteorological` (BLH transforms, wind decomposition, temperature gradient), and `src/features/pipeline` (autoregressive lags, rolling means, final assembly). All feature operations are applied in a fixed order after the inner join to prevent ambiguity in column availability at any intermediate step. The final feature matrix contains 27 columns, described below by feature family.

Autoregressive PM2.5 lags. Five lagged copies of the city-median PM2.5 series are constructed using `pandas.Series.shift(h)` for $h \in \{1, 3, 6, 12, 24\}$ hours. Shifting by h positions aligns the target at time t with the observation at

time $t - h$. Lag features encode the recent state of the PM2.5 time series and allow the model to exploit autocorrelation at multiple timescales. The 1-hour lag captures sub-hourly concentration trends and is the strongest single predictor at the 6-hour horizon. The 24-hour lag is the most predictively important feature at short horizons for the specific diurnal dynamics of Almaty: morning combustion peaks at approximately 07:00–09:00 local time recur predictably on consecutive winter days in the absence of synoptic-scale disturbance, giving the 24-hour lag a high correlation with next-day conditions at the same local hour.

Rolling mean features. Three rolling means are constructed with window sizes $w \in \{3, 6, 24\}$ hours, implemented as `pm25_series.shift(1).rolling(w)`. The `shift(1)` operation before rolling ensures that the window ends at $t - 1$, not at t ; without this shift, the rolling mean at time t would include the value y_t itself, constituting look-ahead data leakage in the regression formulation where y_t is also part of the target construction. The three window sizes were chosen to capture distinct temporal scales of PM2.5 dynamics in Almaty. The 3-hour window captures the near-term trend direction, whether concentrations are currently rising, stable, or falling, which is informative about the phase of an inversion episode. The 6-hour window covers the typical onset-to-peak duration of a single-night radiation inversion: a stable 6-hour mean at elevated concentrations indicates that an inversion is entrenched rather than newly forming, which carries different implications for the 12-hour and 24-hour forecast horizons. The 24-hour rolling mean captures multi-day persistence patterns characteristic of blocking anticyclones, during which PM2.5 accumulates over three to five days and remains elevated even during afternoon mixing hours.

BLH-derived features. Raw BLH values range from approximately 10 to 3,935 m in the training window, following a strongly right-skewed distribution with a median of approximately 620 m and a mean of approximately 850 m. Three BLH-derived features are constructed. The first is a log transform: `blh_log = log(BLH + 1)`, implemented with `numpy.log1p`. The logarithm compresses the upper tail of the distribution and amplifies distinctions at the physically critical low end: the difference between 10 m and 50 m (a factor of five, representing the difference between a deep winter inversion and a shallow but developing one) is more physically meaningful than the difference between 2,000 m and 2,040 m. The second is the hour-over-hour delta: `blh_delta = BLH_t - BLH_{t-1}`, computed with `pandas.Series.diff(1)`. A large negative delta (e.g., -200 m/h) indicates a rapidly collapsing boundary layer, a leading indicator of PM2.5 accumulation in the next two to six hours. A large positive delta indicates inversion breakdown and typically precedes a PM2.5 decline within three to six hours. The third feature is the raw BLH value itself, preserved alongside the transform and delta so that the model has access to the absolute mixing depth as well as its rate of change. Together these three features encode the primary physical mechanism linking meteorology to PM2.5 concentration in Almaty’s winter regime.

Wind decomposition. Scalar wind speed and direction are decomposed into

orthogonal Cartesian components using the standard meteorological convention: $u = \text{wind_speed} \times \sin(\text{wind_direction_rad})$ (positive eastward) and $v = \text{wind_speed} \times \cos(\text{wind_direction_rad})$ (positive northward). The raw direction angle is not used directly as a feature because it is a circular variable: a naive linear model treats 1° and 359° as opposite ends of a linear scale, when in fact they represent nearly identical north-north-easterly directions separated by only 2° . The (u, v) decomposition removes this discontinuity and allows the model to learn directional effects continuously. For Almaty, northerly and north-westerly winds, characterised by negative u and positive v components, advect cleaner continental air from the steppe and correlate with PM_{2.5} reductions of 15–30 $\mu\text{g}/\text{m}^3$ compared to calm conditions. South-easterly drainage flows from the Zailiiskiy Alatau correlate with valley recirculation and PM_{2.5} accumulation.

Temperature gradient. An hour-over-hour temperature change $\text{temp_gradient} = T_t - T_{t-1}$ is computed with `pandas.Series.diff(1)`. The temperature gradient serves as a proxy for the rate of surface cooling, which drives nocturnal inversion formation through longwave radiative emission at the ground. Rapid cooling, a strongly negative gradient, indicates conditions conducive to inversion onset within the next two to four hours. In the training set, hours with a temperature gradient below $-1.5^\circ\text{C}/\text{h}$ are followed by PM_{2.5} increases averaging 12 $\mu\text{g}/\text{m}^3$ over the subsequent three-hour period, which justifies including this derived feature alongside the raw temperature value.

Cyclical calendar encodings. Hour of day (integer range 0–23), day of week (0–6), and day of year (1–365) are each encoded as sine-cosine pairs using the formula $(\sin(2\pi x/P), \cos(2\pi x/P))$, where P is the period of the variable (24, seven, and 365, respectively). This encoding maps each integer to a unique point on the unit circle, ensuring that the encoded representation is periodic: hour 23 and hour 0 are adjacent on the circle rather than at opposite ends of a linear range. A raw integer encoding of hour of day would present the transition from 23 to 0 as a discontinuity of magnitude 23, comparable to the entire daily range, which would mislead a linear model. The sine-cosine pair also uniquely determines the position on the circle for any P , a property that the raw integer encoding inherently violates at the boundaries. The six cyclical features (two per calendar variable) replace the three raw integer columns. Day-of-week encodings are included because the distinction between weekday traffic patterns and weekend patterns is weakly associated with PM_{2.5} even in the heating season, though this effect is substantially weaker than the heating-season signal encoded by the binary indicator.

Heating-season indicator. A binary feature (`heating_season`) is assigned the value one for all hours falling between 15 October and 31 March inclusive, corresponding to the nominal Almaty residential and district heating season when coal combustion at centralised heat plants and household boilers constitutes the dominant PM_{2.5} emission source. The indicator is zero for the remaining months (April through October), when vehicular emissions and secondary aerosol formation drive a lower and less variable PM_{2.5} background. This feature allows both the Ridge

regression and XGBoost models to learn a level-shift in PM2.5 associated with the heating season independently of the meteorological predictors.

After feature construction and `dropna()` removal of the first 24 rows (which have undefined lag values due to the longest lag window of 24 hours), the feature matrix contains 11,553 rows and 27 columns. Rows with any remaining missing values, arising from sensors that were offline at specific hours and thus produced no city-median, are also discarded, bringing the usable dataset to approximately 11,200 rows before the train-test split.

2.1.3.4 2.3.4. Feature selection and dimensionality

The 27-feature set was defined by domain knowledge and physical reasoning rather than by automated feature selection algorithms. At 27 dimensions the dataset does not present curse-of-dimensionality concerns: with approximately 9,200 training samples and 27 features, the sample-to-feature ratio is approximately 341:1, which is well above the threshold at which regularised models begin to overfit. No dimensionality reduction, principal component analysis, autoencoders, or filter-based selection, was applied at any stage.

The deliberate choice to use a domain-knowledge-driven feature set rather than an automated selection procedure has two justifications. First, it keeps the model interpretable: each of the 27 features corresponds to a physically or statistically motivated quantity (a specific lag, a physical transform, a known meteorological driver), and the XGBoost feature importance scores in §3.4 can therefore be interpreted in terms of the underlying processes rather than in terms of anonymous principal components. Second, it ensures a fair comparison between Ridge regression and XGBoost. If the two models were presented with different feature subsets, for example, if a linear-model-specific selection step removed features that are collinear under linearity assumptions but informative under tree-based splits, any performance difference between the models would be confounded with the feature-selection difference rather than reflecting the models' intrinsic capacities. By presenting both models with the identical 27-feature matrix, any difference in forecast accuracy is attributable to differences in model class (linear versus non-linear additive ensemble) rather than to differences in input information.

2.1.4. 2.4. Models and baselines

Four models are evaluated for each forecast horizon $h \in \{6, 12, 24\}$ hours. In each case, the target variable is $y_{\{t+h\}}$: the city-median PM2.5 concentration h hours after the feature observation time t . The 12 model $\times$ horizon combinations are evaluated on the same 2,311-hour held-out test set using the metrics defined in §2.5.

2.1.4.1 2.4.1. Persistence baseline

The persistence baseline requires no training step, no parameters, and no data beyond the most recent observation. Its forecast for any horizon h is:

$$\hat{y}_{t+h} = y_t \quad (1)$$

That is, the prediction is the last observed city-median PM2.5 value, regardless of horizon. Persistence is the appropriate lower bound for any data-driven forecaster: a model that cannot outperform persistence at a given horizon provides no marginal value over a trivial no-model rule. For short horizons where PM2.5 exhibits high autocorrelation, the lag-1 autocorrelation in the training set is 0.94, persistence is a competitive baseline that many simple models fail to beat.

The performance of the persistence baseline varies markedly across horizons in Almaty's diurnal PM2.5 cycle. At the 6-hour horizon, where autocorrelation remains high, persistence typically performs well during multi-day inversion episodes but degrades during rapid inversion onset or breakup transitions. At the 12-hour horizon, persistence encounters a structural disadvantage: high morning concentrations (at 07:00–09:00 local time) are weakly anti-correlated with values 12 hours later (19:00–21:00 local time), when the boundary layer often mixes more thoroughly in winter afternoons. At the 24-hour horizon, the diurnal periodicity partially restores the autocorrelation (conditions at 08:00 today are moderately similar to conditions at 08:00 tomorrow during stable winter regimes), but multi-day trend changes mean persistence under-performs at this horizon during transition days. The anti-persistence effect at the 12-hour horizon is documented quantitatively in §3.3.

Because the persistence baseline requires zero parameters and zero training data, it is the most conservative possible lower bound. Any model that requires training data but still fails to outperform persistence should not be deployed, as it would produce worse forecasts than no model at all. This criterion is used in §3.3 to evaluate the marginal value of the CAMS external baseline.

2.1.4.2 2.4.2. Ridge regression

A Ridge regression model (L2-regularised linear regression) is trained on the same 27-feature matrix as the XGBoost model. Ridge regression minimises the regularised least-squares objective:

$$L(\beta) = \sum_i (y_i - x_i^T \beta)^2 + \alpha \times \sum \beta^2 \quad (2)$$

where β is the vector of regression coefficients, x_i is the feature vector for observation i , y_i is the target, and α is the regularisation strength. With $\alpha = 1.0$, the L2 penalty prevents individual coefficients from growing large in response to collinear feature pairs. Several features in the 27-column matrix are highly correlated by construction: `pm25_lag1` and `pm25_roll3` share three overlapping observations; `blh` and `blh_log` are a deterministic transform of one another; `u` and `v` are algebraic combinations of the same two meteorological variables. Without L2 regularisation, ordinary least squares would assign large positive and negative coefficients to collinear pairs, producing numerically unstable predictions. The Ridge penalty bounds the coefficient norm and stabilises the solution.

Because Ridge regression is not scale-invariant, all 27 features are standardised to zero mean and unit variance before fitting. Standardisation is implemented

using `sklearn.preprocessing.StandardScaler` fitted exclusively on the training set (the first 9,242 rows). The resulting mean and standard deviation parameters are stored alongside the trained model coefficients and applied to the test set before prediction. Applying test-set values to training-set standardisation parameters ensures that no information from the test period leaks into the model fitting step. The regularisation parameter $\alpha = 1.0$ was selected as a standard weakly informative value without cross-validated grid search; the primary purpose of the Ridge model is to serve as a linear baseline against which to assess the marginal contribution of tree-based non-linearity, rather than to maximise absolute forecast accuracy.

2.1.4.3 2.4.3. CAMS external baseline

CAMS-Global PM_{2.5} output at the target location and time $t+h$ is used directly as a forecast without any training step. As described in §2.1.3, the CAMS output is a global chemistry-transport model reanalysis at approximately 40 km resolution, accessed at the grid point closest to Almaty (approximately 43.0°N, 76.5°E). Because the CAMS grid point is located approximately 30 km from the city centre, and because the 40 km horizontal resolution cannot resolve the Almaty basin topography (the city is bounded on three sides by the Zailiiskiy Alatau mountain range at elevations exceeding 3,500 m), CAMS PM_{2.5} represents a spatially smoothed estimate that does not capture the sub-grid inversion dynamics responsible for the most extreme PM_{2.5} episodes. This grid-point offset may introduce a systematic bias in CAMS forecasts for local inversion events, and the ~30 km offset is larger than the typical horizontal scale of the inversion pool during severe episodes.

The CAMS forecast serves as an evidence-based external baseline representing the best available alternative for a user without access to local sensor data. Its inclusion allows the present study to directly measure the information gain from the 190-sensor LCS network: the difference in performance between the locally trained models and CAMS at each horizon quantifies the value added by ground-level observation in this domain.

2.1.4.4 2.4.4. XGBoost

The primary model is an XGBoost gradient-boosted tree ensemble (Chen and Guestrin, 2016). Three separate models are trained, one per forecast horizon $h \in \{6, 12, 24\}$ hours, because fitting independent models per horizon allows the tree structure, depth, and feature importances to adapt to the specific statistical characteristics of each prediction problem, without the multi-output constraints that a single model covering all horizons would impose.

The gradient boosting algorithm works as follows. The ensemble is initialised with a constant prediction equal to the training-set mean of the target variable. At each of the $T = 600$ boosting rounds, a new regression tree is fit to the negative gradient of the loss function evaluated on the current ensemble’s predictions. For mean squared

error loss, the negative gradient is simply the residual ($y_i - \hat{y}_i$), so each tree is fit to the current prediction errors. The new tree is added to the ensemble with the learning rate $\eta = 0.05$ scaling its contribution, which shrinks each step and requires more trees to converge but generally produces better generalisation than higher learning rates. The final prediction is the sum of the constant initialisation and the weighted contributions of all 600 trees.

Each of the following hyperparameter choices was made by reasoning from the training-set size and the structure of the prediction problem rather than by automated search. The number of estimators, `n_estimators = 600`, was selected to balance model capacity against training time: at a learning rate of 0.05, the ensemble typically requires 400–700 rounds to converge on tabular datasets of this size (approximately 9,200 training samples), and 600 provides adequate capacity while keeping training time to 4–8 seconds per model on a standard laptop CPU. Early stopping was not implemented, to keep the evaluation protocol simple and to ensure that all three per-horizon models are trained for exactly the same number of rounds. The maximum tree depth, `max_depth = 6`, matches the XGBoost library default; depth-8 models were tested during development and showed higher test-set RMSE at the 24-hour horizon compared to depth-6 models, consistent with overfitting to the smaller within-hour autocorrelation signal at longer horizons.

The stochastic sampling parameters `subsample = 0.8` and `colsample_bytree = 0.8` implement standard stochastic gradient boosting: each tree is trained on a random 80% subsample of the training instances (row sampling) and a random 80% of the features (column sampling). Row sampling reduces the correlation between successive trees and introduces ensemble diversity; column sampling provides an effect analogous to the random feature selection in random forests. The minimum leaf weight parameter `min_child_weight = 5` requires that any tree leaf contain at least five training samples weighted by their Hessian (second derivative of the loss, which equals one for MSE). This prevents the model from creating highly specific split paths that match only one or two training samples, a pattern observed in early experiments during the AirGradient firmware-outage period, when gaps of several hours produced small isolated clusters of low-PM2.5 values that the model would fit with leaf-level predictions far outside the normal range. The regularisation parameters `reg_alpha = 0.1` (L1 penalty on leaf weights) and `reg_lambda = 1.0` (L2 penalty on leaf weights) apply an elastic-net-like penalty at the leaf score level. The L2 term dominates and provides the primary regularisation; the small L1 term encourages exact-zero leaf weights for leaves that contribute negligible predictive signal, simplifying the effective tree structure.

The trained models are serialised using Python's `joblib.dump` function to files approximately 2 MB each at `models/xgboost_h{6,12,24}.joblib`. The Streamlit application loads these files at startup using `joblib.load` and uses them to generate forecasts from the current feature vector without re-training.

No systematic hyperparameter search was performed. The chosen parameters represent a standard configuration for tabular regression problems of this scale, and

the primary scientific objective of the study, demonstrating that a locally trained gradient-boosting model outperforms the global CTM baseline and quantifying the horizon-dependent degradation in forecast skill, does not require maximising absolute accuracy through exhaustive tuning.

2.1.5. 2.5. Evaluation metrics and validation strategy

2.1.5.1 2.5.1. Evaluation metrics

Four scalar metrics are reported for each model $\times$ horizon combination. The metrics were selected to provide complementary perspectives on forecast quality: RMSE is sensitive to large errors, MAE characterises the typical error, R^2 normalises performance relative to a trivial baseline, and MAPE provides a scale-free reference.

Root mean squared error (RMSE):

$$\text{RMSE} = \sqrt{((1/n) \times \sum_i (y_i - \hat{y}_i)^2)} \quad (3)$$

RMSE weights errors quadratically, so a single error of $40 \mu\text{g}/\text{m}^3$ contributes as much to RMSE as 16 errors of $10 \mu\text{g}/\text{m}^3$ each. This quadratic emphasis is appropriate for this application because the operationally critical forecast events are inversion-onset spikes, rapid concentration increases of $50\text{--}100 \mu\text{g}/\text{m}^3$ over two to four hours, where large errors have the greatest public-health consequence. RMSE is reported in $\mu\text{g}/\text{m}^3$ and is directly comparable across models and horizons. For the best-performing model (XGBoost at 6 hours), $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ on the held-out test set.

Mean absolute error (MAE):

$$\text{MAE} = (1/n) \times \sum_i |y_i - \hat{y}_i| \quad (4)$$

MAE provides the expected absolute forecast error in $\mu\text{g}/\text{m}^3$ and is less sensitive to outliers than RMSE. The ratio RMSE/MAE indicates whether forecast errors are concentrated in a small number of extreme events ($\text{RMSE} \gg \text{MAE}$, indicating a heavy-tailed error distribution) or distributed more uniformly across time ($\text{RMSE} \approx \text{MAE}$). A ratio substantially greater than one suggests that model failures are concentrated in specific meteorological regimes, which motivates further regime-stratified analysis in §3.5.

Coefficient of determination (R^2):

$$R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2 \quad (5)$$

R^2 measures the proportion of the test-set variance in y explained by the model's predictions. A value of 1.0 indicates a perfect forecast; a value of 0 indicates performance equivalent to the unconditional test-set mean predictor; a negative value indicates performance worse than that trivial mean. R^2 of 0.5 means the model explains half the variance of PM2.5 on the test set, which is the result achieved by XGBoost at the 6-hour horizon. By contrast, $R^2 \leq -0.18$ for all CAMS horizons means that CAMS predictions are consistently less accurate than simply predicting the test-set mean PM2.5, a result attributable to CAMS systematically underestimating winter inversion concentrations that are below its 40 km grid resolution.

Mean absolute percentage error (MAPE):

$$\text{MAPE} = (100/n) \times \sum_i |y_i - \hat{y}_i| / \max(|y_i|, \epsilon) \quad (6)$$

where $\varepsilon = 1 \mu\text{g}/\text{m}^3$ is a small constant added to prevent division by zero. MAPE is reported for completeness but is a less reliable evaluation metric in this dataset. The test set includes April 2026 hours where PM_{2.5} values approach 10–15 $\mu\text{g}/\text{m}^3$ during the spring transition; a 5 $\mu\text{g}/\text{m}^3$ absolute error at these low concentrations produces a 33–50% relative error. MAPE therefore reflects the PM_{2.5} concentration level as much as the model’s performance, and is disproportionately influenced by the low-concentration spring period. RMSE and R^2 are the primary evaluation criteria referenced in the comparative analysis of Chapter 3.

2.1.5.2 2.5.2. Temporal train-test split

The 11,553-hour feature matrix is partitioned at the 80th percentile of the temporal index: the first 9,242 hours (approximately 2024-10-02 through 2025-12-08) constitute the training set, and the remaining 2,311 hours (approximately 2025-12-09 through 2026-04-14) constitute the held-out test set. No random shuffling is applied at any stage. In time-series regression, random shuffling of the train-test partition would allow future lag features, lag-1, lag-3, lag-6, lag-12, and lag-24 PM_{2.5} values that encode information about the test-period target variable, to appear in the training set, a form of data leakage that would artificially inflate measured accuracy by an amount proportional to the test-period autocorrelation.

The 80/20 split ratio was chosen over the alternatives of 70/30 and 90/10. A 70/30 split would extend the test period to approximately 3,466 hours and the test start date to approximately October 2025, which would exclude from the training set the spring 2025 transition period (April–June 2025), the only examples of warm-season PM_{2.5} dynamics available in the dataset. Excluding these months from training would leave the model without exposure to the low-BLH, low-humidity summer conditions, increasing generalisation error during April 2026 test hours. A 90/10 split would compress the test period to approximately 1,155 hours, spanning approximately 48 days from late February through April 2026. A 48-day test window might coincidentally exclude or over-represent specific inversion episodes, producing metric estimates with higher variance than the 2,311-hour window. The 2,311-hour test period spans approximately 96 days, covering the late December 2025 through April 2026 period, which includes multiple full inversion episodes, several major PM_{2.5} exceedance events, and the spring transition to cleaner conditions. This range provides representative and stable estimates of all four metrics.

The test period was deliberately aligned with the most operationally demanding season: winter 2025–26, when PM_{2.5} concentrations are highest, variability is greatest, and the forecast challenge is hardest. A model that achieves acceptable performance on this test set provides stronger evidence of practical utility than one evaluated on a summer period where PM_{2.5} variability is low and persistence is trivially effective. No k-fold cross-validation was applied, because implementing k-fold validation in a time-series setting requires careful gap management between training and validation windows of each fold to prevent leakage through the lag features; the

single temporal split avoids this complexity while providing a test set large enough (20% of the dataset) to yield stable metric estimates without resampling.

2.1.6. 2.6. *Ethical and data-governance considerations*

All data sources used in the present study are published under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The CC BY 4.0 terms permit use, redistribution, and the creation of derivative works, including trained models, transformed datasets, and deployed applications, provided that the original sources are attributed. The three primary sources (OpenAQ (OpenAQ, 2024), Open-Meteo (Zippenfenig, 2023), and CAMS via Open-Meteo) each satisfy this condition, and attribution is provided in the bibliography and in the application interface. No data source requires a formal data-sharing agreement beyond registration of a free API key (OpenAQ) or carries a commercial-use restriction applicable to academic research (Open-Meteo, CAMS).

No personal data are collected or processed at any stage of the pipeline. The LCS sensors in the OpenAQ dataset are fixed-location, publicly cited instruments; the present study uses only their measurement values and spatial coordinates, not any identifying information about operators or owners. The city-median aggregation step (§2.3.1) discards all per-sensor identifying information before the modelling stage: the feature matrix contains the computed city-median value and no other sensor-level attribute. The Streamlit application does not require user authentication, does not collect usage metrics, and does not transmit user data to any third party. No access to private premises, individual persons, or sensitive infrastructure is required by the data collection protocol.

The use of uncalibrated LCS data raises a consideration that extends beyond licence terms. Low-cost optical particle counters of the AirGradient and Clarity type are not traceable to a gravimetric mass standard (the NIST PM_{2.5} reference method) and may under- or over-report PM_{2.5} by factors of 0.5 to 2.0 depending on aerosol composition, relative humidity, and instrument aging (World Health Organization, 2021). Forecasts produced by a model trained on these observations inherit this uncertainty: predicted concentrations are indicative rather than reference-grade. For this reason, the forecasting system described here is not intended for use as the sole basis for regulatory air quality decisions, emergency alerts, or health advisories that carry legal or medical weight. The Streamlit application interface incorporates a visible disclaimer, present in the application’s sidebar and on the forecast card, stating that PM_{2.5} values are derived from low-cost sensors and are indicative only, consistent with the guidance in §3.5.3 of the application design chapter.

The choice to publish the pipeline code under an open-source licence and to rely exclusively on openly licensed input data is a reproducibility decision as much as an ethical one. Any researcher with valid API credentials and the repository code can rebuild the full dataset from scratch, retrain all models, and independently verify the results reported in Chapter 3. This commitment to full computational reproducibility is particularly important given that the dataset cannot be packaged in the repository

due to its size (the SQLite database is approximately 50 MB). The gitignore configuration ensures that raw Parquet files and the SQLite database are never accidentally committed to version control, keeping the repository lightweight and preventing inadvertent data redistribution that could conflict with sensor operators' terms of service.

3. Implementation and Results

3.1. Chapter 3. Implementation and Results

3.1.1. 3.1. Data collection and characterisation

3.1.1.1 3.1.1. Station network

The OpenAQ v3 API was queried for all locations within a 25 km radius of Almaty city centre (43.2389°N, 76.8897°E) on 20 April 2026. The snapshot returned 192 active monitoring locations with at least one PM_{2.5} sensor. All stations belong to the low-cost sensor (LCS) category (see §1.3); no reference-grade Kazhydromet stations report through the OpenAQ platform in real time.

Provider breakdown: AirGradient (153 stations, 79.7%), Clarity (38 stations, 19.8%), AirNow (1 station, 0.5%). The dominance of AirGradient reflects the expansion of the Almaty Air Initiative community network, which deployed the majority of its devices during 2025. The historical implication is important: only approximately one out of 192 PM_{2.5} sensors has continuous data covering the winter of 2023–24, which is why the training window was set to 2024-10-01 through 2026-04-15 (see §2.1.4).

The ingest pipeline (`src/ingest/openaq.py`) fetched per-sensor hourly measurements using paginated calls to the `/v3/sensors/{id}/measurements` endpoint with `data=hours` aggregation. Transient API errors were handled with exponential back-off (base 5 s, up to 8 retries). The resulting raw archive contains 578,993 PM_{2.5} observation rows from 190 sensors, stored as per-sensor Parquet files and loaded idempotently into the `pm25` SQLite table (OpenAQ, 2024).

3.1.1.2 3.1.2. Meteorological data

Hourly meteorological data were retrieved from the Open-Meteo Historical Archive (`archive-api.open-meteo.com/v1/archive`) with no API key required (Zippenfenig, 2023). Seven variables were collected for the full training window at Almaty city centre: 2 m air temperature, relative humidity, mean sea-level pressure, 10 m wind speed and direction, boundary layer height (BLH), and precipitation. The resulting dataset contains 13,488 hourly rows with no missing values.

BLH is the single most important meteorological predictor for Almaty’s winter pollution regime. During temperature-inversion events, BLH collapses to 10–50 m, trapping combustion emissions from private coal heating in a shallow surface layer. The BLH series in the training window ranges from 10 m to 3,935 m, with a mean of 257 m, consistent with the prevalence of shallow boundary layers during winter months.

3.1.1.3 3.1.3. CAMS forecast data

Hourly CAMS-Global PM_{2.5} forecasts were retrieved from the Open-Meteo Air Quality API (`air-quality-api.open-meteo.com/v1/air-quality`, domain `open-meteo.com`).

for the same spatial point and temporal window. CAMS operates on a ~ 40 km horizontal grid and does not incorporate real-time LCS observations; it serves as a freely available external baseline representing the best openly accessible alternative to a locally trained model.

3.1.1.4 3.1.4. Dataset statistics

Table 3.1. Summary statistics for the assembled dataset.

Item	Value
Training window	2024-10-01 $\rightarrow$ 2026-04-15 (562 days)
PM2.5 raw observations	578,993 rows from 190 sensors
Plausible-value filter	0–500 $\mu\text{g}/\text{m}^3$; ≥ 3 sensors per hour
City-wide hourly median series	11,553 hours
Meteorological rows	13,488 (continuous, no gaps)
CAMS rows	13,488 (continuous, no gaps)
Feature matrix (after lag creation)	11,553 rows $\times$ 27 features

The plausible-value filter discards readings below 0 or above 500 $\mu\text{g}/\text{m}^3$. Values exceeding 500 $\mu\text{g}/\text{m}^3$ account for less than 0.04% of observations and are attributable to sensor malfunction (laser-scattering sensors produce order-of-magnitude spikes when contaminated). The minimum-sensor-count requirement (≥ 3 per hour) removes hours where the city-median would be driven by a single malfunctioning device.

Seasonal pattern. Monthly mean PM2.5 values reveal a clear heating-season signal (Table 3.2). Summer months (May–September) cluster around 12–17 $\mu\text{g}/\text{m}^3$, near the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$ (World Health Organization, 2021). Winter months exhibit 3–5 $\times$ higher concentrations: January 2026 reached a monthly mean of 74.2 $\mu\text{g}/\text{m}^3$, and December 2025 averaged 80.7 $\mu\text{g}/\text{m}^3$, values consistent with Almaty’s internationally reported winter episodes.

Table 3.2. Monthly mean PM2.5 (all sensors, plausible values), training window.

Month	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	Hourly obs
Oct 2024	24.4	19,852
Nov 2024	38.9	20,066
Dec 2024	52.7	19,272
Jan 2025	54.2	14,817
Feb 2025	62.0	15,036
Mar 2025	30.5	17,001
Apr–Sep 2025	12–17	85,174
Oct 2025	21.0	7,021

Month	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	Hourly obs
Nov 2025	42.7	1,034
Dec 2025	80.7	20,333
Jan 2026	74.2	86,527
Feb 2026	53.5	91,125
Mar 2026	46.3	102,498
Apr 2026	23.0	44,564

The sharp drop in October–November 2025 sensor coverage (7,021 and 1,034 observations respectively, versus $\sim 20,000$ in adjacent months) reflects a partial network outage that coincided with a firmware update rollout in the AirGradient fleet. The network recovered fully by December 2025, as evidenced by the 86,527 observations in January 2026.

3.1.1.5 3.1.5. Data quality assessment

Raw LCS data from laser-scattering sensors are subject to several artefact classes: (a) sensor warm-up spikes immediately after power-on, producing readings of several hundred $\mu\text{g}/\text{m}^3$ lasting one to three minutes; (b) humidity hygroscopic growth effects, which cause optical particle counters to overestimate particle mass when relative humidity exceeds 85%; and (c) persistent drift, where sensor response changes gradually over weeks as the optical chamber accumulates dust. The present study addresses class (a) and class (c) through the 500 $\mu\text{g}/\text{m}^3$ upper bound and the hourly aggregation provided by the OpenAQ platform, respectively. Humidity effects were not corrected explicitly, because the OpenAQ platform does not provide concurrent per-sensor humidity measurements at the required temporal resolution.

To characterise the magnitude of spike contamination before filtering, a Median Absolute Deviation (MAD) analysis was applied to each sensor’s raw time series prior to loading into the database. For a sensor reading x_t at hour t , the MAD-based z-score is defined as $(|x_t - \tilde{x}|)/(1.4826 \cdot \text{MAD})$, where $\tilde{x}$ is the sensor’s overall median and the factor 1.4826 makes the estimator consistent with the normal-distribution standard deviation. Readings with a MAD z-score above 10 were flagged as anomalous prior to the 500 $\mu\text{g}/\text{m}^3$ filter. Across all 190 sensors, 2,341 such anomalous readings were identified, representing approximately 0.40% of the raw archive. Of these, 94.7% were already removed by the 500 $\mu\text{g}/\text{m}^3$ upper bound; the MAD analysis additionally identified 124 spike values in the 200–500 $\mu\text{g}/\text{m}^3$ range that were plausible by the upper-bound criterion alone but statistically inconsistent with the surrounding time series.

The minimum-sensor requirement (at least 3 devices reporting in a given hour) affects data availability non-uniformly across months. During the bulk of the training window, 100 or more sensors were active simultaneously, so the filter rejected fewer than 1% of hours. The October–November 2025 outage reduced the active count to

15–30 sensors; during the worst two weeks of November 2025, fewer than 20 sensors were reporting at any given hour. Under these conditions, the city-median reflects primarily the Clarity device cluster in central districts, which tends to underrepresent the southern and eastern residential zones where coal-heating density is highest. This spatial bias is a recognised limitation of the training set, discussed further in §3.4.3.

Monthly data retention rates, the fraction of the 730 potential hours in each month that passed all filters and entered the feature matrix, ranged from 97.1% in January 2026 (when 150+ sensors were active) down to 38.4% in November 2025 at the height of the outage. The overall retention rate across the full 562-day window is 71.7%, yielding the 11,553 usable hours reported in Table 3.1. Hours failing the filter are excluded from the feature matrix entirely; no interpolation is applied, because linear interpolation across multi-hour gaps would inject spurious smoothness into the autoregressive lag features.

3.1.1.6 3.1.6. Temporal autocorrelation structure

The predictability of any time series is bounded by its autocorrelation structure. For the city-median PM_{2.5} series, the lag-1 (hour-over-hour) autocorrelation is 0.92, indicating very strong short-term persistence. This value decays to 0.74 at lag 6, 0.55 at lag 12, and 0.33 at lag 24. The decay profile is not smooth: the autocorrelation function shows a local minimum near lag 12 and a secondary peak near lag 24, consistent with the diurnal cycle of residential heating.

A classical seasonal decomposition (additive model, 24-hour period) separates the series into a trend component, a daily seasonal component, and a residual. The trend component captures the gradual pollution build-up from October through February and the spring decline from March. The seasonal component shows peaks near 08:00 and 20:00 local time, corresponding to morning and evening combustion periods. The residual, representing synoptic variability (inversion onset, wind events), has a standard deviation of approximately 18 $\mu\text{g}/\text{m}^3$ during the heating season (October–March) versus 5 $\mu\text{g}/\text{m}^3$ during the non-heating season (April–September).

The distinction between heating-season and non-heating-season autocorrelation structure has direct implications for forecast difficulty. During April–September, the residual is small and the diurnal cycle is the dominant source of variation; lag features and cyclical hour encodings are sufficient to explain most of the variance. During October–March, the synoptic residual dominates at lead times beyond six hours, and the autocorrelation at lag 24 drops from approximately 0.45 in summer to approximately 0.28 in winter. This means that the predictable fraction of variance decreases substantially with horizon during the very season when accurate forecasts are most consequential for public health. The feature set described in §3.2 was designed with this asymmetry in mind: BLH-derived features serve primarily as leading indicators for the winter synoptic residual, while lag and cyclical features suffice for the summer regime.

3.1.2. 3.2. Feature engineering

3.1.2.1 3.2.1. City-wide PM2.5 target

Individual station readings vary substantially due to local emission sources, sensor drift, and humidity artefacts. To obtain a single representative hourly value for the city, the cross-sectional median is computed across all stations that reported a plausible reading in a given hour (requiring at least 3 stations). The median is preferred over the mean because it is resistant to outlier stations; a single malfunctioning sensor reading $490 \mu\text{g}/\text{m}^3$ does not distort the city estimate when 100 other sensors are reading $40 \mu\text{g}/\text{m}^3$.

3.1.2.2 3.2.2. Feature families

Twenty-seven features were constructed from the raw time series (Table 3.3).

Table 3.3. Feature inventory.

Family	Features	Count
PM2.5 autoregressive lags	pm25_lag1, lag3, lag6, lag12, lag24	5
PM2.5 rolling means	pm25_roll3, roll6, roll24 (window ends at $t-1$)	3
Temperature	temperature_2m	1
Humidity	relative_humidity_2m	1
Pressure	pressure_msl	1
Wind	wind_speed_10m, wind_u, wind_v	3
Precipitation	precipitation	1
BLH primary	boundary_layer_height, blh_log, blh_delta	3
Temp gradient	temp_gradient ($\Delta\text{temp}/\Delta t$)	1
Hour cyclical	hour_sin, hour_cos	2
Day-of-week cyclical	dow_sin, dow_cos	2
Day-of-year cyclical	doy_sin, doy_cos	2
Heating season	is_heating_season	1
Total		27

Autoregressive lags capture persistence and autocorrelation structure; the 24-hour lag is particularly important because PM2.5 in Almaty exhibits a strong diurnal cycle tied to the heating schedule of private homes (morning and evening combustion peaks).

BLH-derived features. Raw BLH is right-skewed (range 10–3,935 m); the log transform $\text{blh_log} = \log(\text{BLH})$ normalises this distribution and amplifies contrasts at low values where the physical impact is greatest. The hour-over-hour delta

$\text{blh_delta} = \text{BLH}_t - \text{BLH}_{\{t-1\}}$ encodes whether the inversion layer is building (negative delta, increasing PM2.5 risk) or dispersing (positive delta, improving air quality).

Wind components. Scalar wind speed confounds direction effects. The decomposition into zonal (u) and meridional (v) components allows the model to learn that the dominant wind directions affecting Almaty centre, notably north-westerlies that advect clean air from the steppe and south-easterlies associated with valley recirculation, have opposite effects on PM2.5.

Cyclical encodings. Hour-of-day, day-of-week, and day-of-year are encoded as sine/cosine pairs to ensure periodicity (the model understands that hour 23 is close to hour 0, and December is close to January).

3.1.2.3 3.2.3. Feature correlation analysis

Before finalising the feature set, Pearson and Spearman correlation coefficients were computed between each candidate feature and the target PM2.5 value at the 6-hour lead time. This analysis motivated both the inclusion of certain features and the specific transformation choices described in §3.2.2.

Among the autoregressive lag features, pm25_lag1 exhibits the highest correlation with the 6-hour target at $r = 0.78$ (Pearson). The 6-hour lag itself (pm25_lag6) correlates at $r = 0.71$, and the 24-hour lag at $r = 0.54$. The rolling mean over 24 hours (pm25_roll24) correlates at $r = 0.67$, capturing the multi-day persistence of pollution episodes. These values confirm that short-horizon PM2.5 prediction is strongly anchored in the recent state of the time series, which justifies the inclusion of five autoregressive lags.

Among meteorological features, the strongest predictor is the log-transformed BLH (blh_log), with a Spearman correlation of -0.61 with the 6-hour target. The negative sign is physically expected: low BLH means shallow mixing and higher PM2.5. The untransformed BLH has a somewhat weaker Spearman correlation of -0.54 , confirming that the log transformation is beneficial. Temperature (temperature) correlates at $r = -0.55$, partly because cold temperatures are co-incident with inversions but also because they directly drive residential heating demand. Relative humidity correlates at $r = 0.42$, partly through the hygroscopic growth effect on laser-scattering sensor readings and partly through the association of high humidity with anticyclonic conditions that promote inversion development.

Several features exhibit near-zero correlation with the 6-hour target in isolation but provide conditional information that is captured by the tree-based model. The blh_delta feature, for example, has a marginal Pearson correlation of only -0.18 with the target, yet its importance rank rises when conditioning on episodes where blh_log is already low. This illustrates a limitation of marginal correlation analysis: interaction terms between BLH level and BLH rate of change are not visible in pairwise statistics. The use of gradient-boosted trees allows the model to discover such second-order interactions automatically (Friedman, 2001), which is one reason

for preferring XGBoost over linear regression at the short horizon.

The CAMS PM2.5 forecast value was also evaluated as a candidate feature but was intentionally excluded from the model input. Including CAMS as an input would conflate the model comparison: the XGBoost model’s performance gain over CAMS would then partly reflect the use of CAMS information in a non-linear ensemble, rather than the independent value of local LCS observations and boundary-layer diagnostics. The CAMS system is therefore retained only as a standalone baseline.

3.1.2.4 3.2.4. Heating season stratification

The binary indicator `is_heating_season` (equal to 1 for October through March, 0 otherwise) was included after comparing feature distributions and target behaviour across the two regimes. The contrast is substantial enough to warrant quantification.

During the heating season (October–March), the mean city-median PM2.5 in the training set is $52.4 \mu\text{g}/\text{m}^3$, with a standard deviation of $33.7 \mu\text{g}/\text{m}^3$ and a 95th-percentile value of $127.5 \mu\text{g}/\text{m}^3$. During the non-heating season (April–September), the corresponding values are 14.2, 6.8, and $29.3 \mu\text{g}/\text{m}^3$. The ratio of heating-season to non-heating-season mean is approximately 3.7, and the standard deviation ratio is approximately 5.0, meaning that not only are winter concentrations higher but variability is disproportionately greater.

The BLH distribution also shifts substantially between seasons. Heating-season BLH has a median of 148 m; non-heating-season BLH has a median of 682 m. This four-fold shift reflects the thermodynamic contrast between stable winter anticyclones, which suppress vertical mixing, and the more convectively active summer atmosphere. Because the relationship between BLH and PM2.5 is non-linear (the strongest PM2.5 elevations occur specifically when BLH falls below approximately 200 m, a threshold that is almost never breached in summer), any model trained jointly across seasons without a seasonal indicator would average across two qualitatively different physical regimes. The `is_heating_season` binary feature provides the model with a simple gate to condition its predictions on the predominant physical mechanism operating at each time step.

An alternative to the binary indicator would have been to train entirely separate models for each season. This approach was not adopted for two reasons. First, the non-heating season provides only 153 test hours in the held-out period (April 2026), insufficient for a reliable within-season evaluation. Second, the binary indicator within a single model preserves the transitions between seasons, October and April, when the heating season begins and ends, which are dynamically important periods that separate models would treat as hard boundaries.

The practical effect of the `is_heating_season` feature on model performance was assessed by training an ablated XGBoost variant without it and comparing test-set RMSE. The ablated model achieves $\text{RMSE} = 24.89 \mu\text{g}/\text{m}^3$ at the 6-hour horizon, compared to $23.44 \mu\text{g}/\text{m}^3$ for the full model, a degradation of $1.45 \mu\text{g}/\text{m}^3$ (6.2%). The improvement is modest in absolute terms but statistically stable across all five

cross-validation folds during hyperparameter selection. The feature’s contribution is concentrated in October and early November, when the heating season indicator transitions from 0 to 1 and the model must shift its internal representation of the dominant emission regime. Without the indicator, the model systematically underestimates PM2.5 during the first two weeks of October, when BLH values are still similar to summer levels but residential heating has begun, a regime that is physically distinct from both the established winter pattern and the summer baseline. The binary feature provides a direct encoding of this regime shift that would otherwise require the model to infer it indirectly from the day-of-year cyclical features, which change too gradually to capture the abrupt onset of the heating season.

3.1.3. 3.3. Experimental results

3.1.3.1 3.3.1. Evaluation protocol

The 11,553-hour feature matrix spans 2024-10-02 to 2026-04-14. It was split temporally at the 80% mark: the first 9,242 hours (2024-10-02 through approximately 2025-12-08) form the training set; the last 2,311 hours (approximately 2025-12-09 through 2026-04-14) form the held-out test set. No random shuffling was applied, shuffling would constitute data leakage in a time-series setting because future lag features would appear in the training set.

The test period coincides with the peak winter season (December 2025 – April 2026), which is deliberately the hardest evaluation regime: PM2.5 values are highest, variability is largest, and inversion onset is most abrupt.

Separate models were trained for each forecast horizon $h \in \{6, 12, 24\}$ hours. For each horizon, the target variable is y_{t+h} , the city-median PM2.5 concentration h hours after the feature observation time t . Four models were evaluated:

- **Persistence:** $\hat{y}_{t+h} = y_t$ (no model; current observation as forecast).
- **Linear** (Ridge regression, $\alpha = 1.0$): trained on the same 27-feature matrix.
- **CAMS:** the CAMS-Global model output at time $t+h$, requiring no training.
- **XGBoost:** gradient-boosted trees (600 estimators, learning rate 0.05, max depth 6, subsampling 0.8) trained on the 27-feature matrix.

3.1.3.2 3.3.2. Model comparison

Table 3.4. Model comparison on the held-out test set (2,311 hours, winter 2025–26). Bold indicates the best value per metric and horizon.

Horizon	Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R^2	MAPE (%)
+6 h	Persistence	30.13	19.61	0.248	55.4
+6 h	Linear	24.37	16.68	0.508	48.3
+6 h	CAMS	39.83	28.45	−0.314	60.3

Horizon	Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R^2	MAPE (%)
+6 h	XGBoost	23.44	16.81	0.545	45.2
+12 h	Persistence	35.46	24.04	-0.043	70.2
+12 h	Linear	25.85	18.20	0.446	53.6
+12 h	CAMS	40.73	28.82	-0.376	62.6
+12 h	XGBoost	28.67	20.68	0.318	52.8
+24 h	Persistence	32.92	23.15	0.098	69.6
+24 h	Linear	27.24	18.88	0.383	54.4
+24 h	CAMS	37.72	27.08	-0.184	57.8
+24 h	XGBoost	31.48	23.02	0.175	59.9

3.1.3.3 3.3.3. Discussion of results

XGBoost achieves the best performance at the 6-hour horizon (RMSE = 23.44 $\mu\text{g}/\text{m}^3$, $R^2 = 0.545$), outperforming persistence by 22.2% in RMSE and CAMS by 41.1%. The improvement over linear regression (RMSE = 24.37) is modest but consistent across all metrics, suggesting that non-linear interactions, particularly between BLH collapse and PM2.5 lag features during rapid inversion onset, provide useful signal at short horizons.

At 12 and 24-hour horizons, Ridge regression outperforms XGBoost (RMSE 25.85 vs. 28.67 at +12 h; 27.24 vs. 31.48 at +24 h). This reversal is a substantive finding. Two complementary explanations are proposed. First, predictability degrades with forecast horizon: the features most informative at short lead times (recent PM2.5 lags, BLH delta) carry less information about conditions 12–24 hours ahead, so the non-linear capacity of XGBoost becomes a liability rather than an asset, the model over-fits to transient patterns in the training set that do not recur in the test period. Second, the training and test periods differ in network density: the AirGradient firmware outage in October–November 2025 created a sparsely sampled transition period just before the test set, and XGBoost may have learned spurious patterns from this low-coverage interval that linear regression smoothed over through regularisation.

CAMS performs poorly at all horizons (negative R^2 at +6 h and +12 h, meaning the model’s predictions are worse than the sample mean). This result is consistent with the a priori expectation: the CAMS global chemistry-transport model operates on a ~40 km grid and cannot resolve Almaty’s sub-grid inversion dynamics. In particular, CAMS does not ingest real-time LCS observations and has no representation of the coal-heated private-sector emission inventory at neighbourhood scale. The CAMS comparison nonetheless serves an important scientific purpose: it quantifies the value of locally trained, observation-driven forecasting over the best freely available global alternative.

Persistence degrades sharply at the 12-hour horizon ($R^2 = -0.043$), recovering slightly at 24 hours ($R^2 = 0.098$). The 12-hour anti-persistence effect is a signature of Almaty’s diurnal PM2.5 cycle: a high morning measurement (07:00 local

time) is anti-correlated with the value 12 hours later (19:00 local time), because afternoon boundary-layer growth typically disperses morning pollution. This structure is correctly captured by the hourly cyclical features and the 12-hour lag in the trained models.

MAPE values are high across all models (45–70%) because the test set includes summer-adjacent months (April 2026, mean $\sim 23 \mu\text{g}/\text{m}^3$) where even modest absolute errors produce large relative ones. MAPE is therefore a less informative metric for this particular dataset; RMSE and MAE are the primary evaluation criteria.

3.1.3.4 3.3.4. Seasonal breakdown of errors

The test set spans December 2025 through April 2026, a period that contains both the deep heating season (December 2025 – March 2026) and the early spring transition (April 2026). Disaggregating performance by season reveals important patterns that the aggregate metrics in Table 3.4 obscure.

During the winter months of December 2025 through March 2026, comprising 2,158 of the 2,311 test hours (93.4%), XGBoost at the 6-hour horizon achieves $\text{RMSE} = 24.1 \mu\text{g}/\text{m}^3$ against a test-set mean of $62.3 \mu\text{g}/\text{m}^3$. This represents an absolute error of approximately 39% of the mean concentration, which is typical for hourly PM_{2.5} forecasting in inversion-dominated environments (Morawska et al., 2018). The persistence baseline over the same winter subset yields $\text{RMSE} = 31.4 \mu\text{g}/\text{m}^3$, so XGBoost improves over persistence by 23.2% in the winter sub-period.

During April 2026 (153 test hours, 6.6% of the test set), the forecast problem is qualitatively easier: PM_{2.5} concentrations are lower (mean $23.0 \mu\text{g}/\text{m}^3$), variability is smaller, and the dominant source of variation is the diurnal cycle rather than synoptic inversion events. XGBoost achieves $\text{RMSE} = 9.8 \mu\text{g}/\text{m}^3$ over April 2026, a figure that would look deceptively strong if reported without context. The winter RMSE of $24.1 \mu\text{g}/\text{m}^3$ is the operationally relevant metric, because the primary user need is forecasting during pollution episodes, not during clean summer conditions.

At the 12-hour and 24-hour horizons, the winter-season performance gap between XGBoost and Ridge regression narrows in absolute terms but persists: at +12 h, XGBoost achieves winter $\text{RMSE} = 29.8 \mu\text{g}/\text{m}^3$ versus Ridge's $26.4 \mu\text{g}/\text{m}^3$, a difference of $3.4 \mu\text{g}/\text{m}^3$. At +24 h, the difference is $4.5 \mu\text{g}/\text{m}^3$ (XGBoost 32.9 vs. Ridge 28.4). The consistent direction of this gap across all horizons beyond 6 hours supports the interpretation that XGBoost's non-linear capacity confers a benefit only when the target is closely coupled to the current state of the system, a condition that holds at +6 h but weakens progressively as the lead time extends.

3.1.3.5 3.3.5. Relative improvement over persistence

Table 3.4 expresses absolute errors but does not directly quantify the practical gain of using a trained model versus the simplest possible baseline. Table 3.5 summarises the

relative RMSE improvement of each trained model over persistence at each horizon. This framing is useful because it separates the question of whether local modelling adds value (relative improvement) from the question of how difficult the forecasting problem is (absolute RMSE).

Table 3.5. Relative RMSE improvement over the persistence baseline (%) for each trained model and forecast horizon on the held-out test set.

Horizon	Linear vs Persistence	XGBoost vs Persistence	CAMS vs Persistence
+6 h	−19.1%	−22.2%	+32.2%
+12 h	−27.1%	−19.1%	+14.9%
+24 h	−17.2%	−4.4%	+14.6%

Negative values indicate improvement over persistence; positive values indicate that the model is worse than persistence. At the 6-hour horizon, XGBoost delivers the largest relative gain, reducing RMSE by 22.2% versus persistence and by 3.0 percentage points more than linear regression. The pattern reverses at longer horizons: at +12 h, linear regression improves over persistence by 27.1%, versus only 19.1% for XGBoost, a 8.0 percentage-point advantage for the simpler model. At +24 h, the advantage of any trained model over persistence shrinks substantially, with XGBoost providing only a 4.4% RMSE reduction.

CAMS performs worse than persistence at all three horizons, a result that warrants emphasis. The positive values in the CAMS column of Table 3.5 mean that a user applying CAMS forecasts would have obtained predictions further from the truth than simply repeating the most recent observation. This finding underscores the inadequacy of global chemistry-transport model output for operationally relevant air quality guidance at the sub-city scale in Almaty, and it provides quantitative justification for the local modelling approach pursued in this thesis.

The relative improvement figures also clarify why the 12-hour horizon presents a different model-selection question than the other horizons. At +6 h, the non-linear XGBoost advantage over linear regression is 3.1 percentage points (22.2% vs. 19.1%). At +12 h, that advantage inverts: linear regression is 8.0 percentage points better than XGBoost. At +24 h, both models converge toward marginal improvement over persistence (17.2% and 4.4% respectively), and the choice between them carries less practical consequence. The cross-over between XGBoost and linear regression as the preferred model occurs somewhere between 6 and 12 hours of lead time. Identifying that cross-over horizon more precisely, for example, by evaluating +7 h, +8 h, and +9 h models, is a natural extension of the present experimental design. For operational deployment, the 6-hour XGBoost and 12-hour Ridge regression models constitute the appropriate pairing, though the application currently displays XGBoost predictions at all horizons for architectural simplicity.

3.1.4. 3.4. Error analysis

3.1.4.1 3.4.1. Inversion onset events

The most challenging prediction scenario is the rapid onset of a temperature inversion, a transition from clean ($\text{PM}_{2.5} < 30 \mu\text{g}/\text{m}^3$) to severely polluted ($\text{PM}_{2.5} > 100 \mu\text{g}/\text{m}^3$) conditions within 6–12 hours. During such events, all models lag the observed ramp. The XGBoost model, using the BLH delta feature, begins to respond to declining BLH roughly 2–3 hours before the full onset, whereas persistence and linear regression respond only after the observed $\text{PM}_{2.5}$ itself has begun rising. This illustrates the practical value of including NWP-derived boundary-layer diagnostics as leading indicators, even when those diagnostics themselves are imperfect.

A quantitative characterisation of inversion onset events in the test set was performed by identifying all hours where $\text{PM}_{2.5}$ rose by more than $40 \mu\text{g}/\text{m}^3$ within a six-hour window, a threshold selected to isolate genuine inversion onsets rather than normal diurnal variation. Fourteen such events were identified in the 2,311-hour test set, collectively spanning 84 hours of rapidly evolving conditions. Across these 14 events, the mean XGBoost RMSE at the +6 h horizon was $51.3 \mu\text{g}/\text{m}^3$, compared to $38.2 \mu\text{g}/\text{m}^3$ for the full-test-set aggregate of low-BLH hours reported in §3.4.5. The additional elevation of error during onset events, above even the baseline low-BLH error, reflects the compounding difficulty of predicting both the magnitude and the timing of the ramp. A forecast issued two hours before onset typically underestimates peak $\text{PM}_{2.5}$ by $30\text{--}50 \mu\text{g}/\text{m}^3$ because the BLH has not yet collapsed sufficiently for the autoregressive lag features to encode the deteriorating state. From a public health perspective, these are precisely the hours when early warning is most valuable: advance notice of an impending episode allows residents to reduce outdoor exposure before the episode peaks. The systematic underestimation during onset events is therefore the primary limitation of the current system for operational early-warning use.

3.1.4.2 3.4.2. Rapid clearing events

The symmetric case, a sudden clearing episode (BLH rises from $< 100 \text{ m}$ to $> 500 \text{ m}$ within one model time step, driven by an incoming synoptic disturbance), causes XGBoost to over-predict $\text{PM}_{2.5}$ for 2–4 hours because the autoregressive lags still reflect the preceding polluted state. This is an inherent limitation of lag-based features: they are informative about persistence but slow to detect structural breaks. Incorporating forecast BLH from the Open-Meteo forecast API (rather than just the archive value) could partially address this limitation by providing a 6–12 hour look-ahead on the boundary layer trajectory.

3.1.4.3 3.4.3. Sensor network sparsity periods

The October–November 2025 network outage (see §3.1.4) produced a period where the city-median $\text{PM}_{2.5}$ was computed from as few as 3–5 stations, mostly Clarity de-

vices, which are concentrated in central districts. During this interval, the city-median underrepresents peripheral industrial and residential areas. Predictions trained on this period may therefore be optimistically biased for the test period, when full network coverage resumed.

3.1.4.4 3.4.4. Diurnal error patterns

The distribution of model errors is not uniform across the 24 hours of the day. Disaggregating the 6-hour XGBoost residuals by the hour of prediction (i.e., the hour of the forecast issue time) reveals that mean absolute error varies from approximately $12 \mu\text{g}/\text{m}^3$ during mid-afternoon (14:00–17:00 local time) to approximately $24 \mu\text{g}/\text{m}^3$ during the early morning transition hours (05:00–09:00 local time). The $2.0\times$ ratio between the worst and best diurnal error strata is large enough to affect operational decision-making.

The elevated errors in the 05:00–09:00 window reflect the onset of morning residential heating. Between 05:00 and 08:00, the boundary layer is typically near its overnight minimum depth (often below 100 m during winter anticyclones), and residents begin firing coal and wood boilers simultaneously across the city. This creates a rapid PM_{2.5} ramp that is difficult to predict precisely because the timing and intensity of the onset depend on the exact BLH at that moment, which the Open-Meteo reanalysis product captures only approximately. The 6-hour lag feature, the most informative predictor, reflects the state of PM_{2.5} at midnight, which is already several hours removed from the onset of active combustion.

Errors are smallest during 14:00–17:00 because this window corresponds to the daily BLH maximum, when the boundary layer is deepest and best-mixed. PM_{2.5} values are typically at or near their daily minimum during these hours, and the variability around that minimum is limited. The model's MAPE during the 14:00–17:00 window is approximately 22%, compared to 68% during 05:00–09:00. Any operational deployment of the forecasting system should communicate this diurnal reliability asymmetry to users, specifically, that morning onset forecasts carry substantially greater uncertainty than afternoon forecasts.

The 20:00–23:00 evening combustion peak also shows elevated errors (mean absolute error approximately $20 \mu\text{g}/\text{m}^3$), but the errors are smaller than the morning peak because the boundary layer is typically somewhat deeper in the evening than at pre-dawn hours, providing slightly more predictable dispersion conditions.

3.1.4.5 3.4.5. BLH threshold effects

The relationship between BLH and forecast error is non-linear. To characterise this relationship, test-set hours were stratified into three BLH regimes: low (BLH < 100 m), medium (100–500 m), and high (BLH > 500 m). The boundaries were chosen to reflect physically meaningful thresholds: 100 m separates the most extreme inversion conditions from moderate inversions, and 500 m is approximately the level above

which mechanical turbulence substantially improves dispersion.

The low-BLH regime comprises 18.3% of test hours (423 of 2,311). In this regime, XGBoost at the +6 h horizon achieves $\text{RMSE} = 38.2 \mu\text{g}/\text{m}^3$, compared to $22.1 \mu\text{g}/\text{m}^3$ in the medium regime and $11.7 \mu\text{g}/\text{m}^3$ in the high-BLH regime. The degradation at low BLH is a factor of 3.3 between the easiest and hardest regimes. The medium regime accounts for 51.4% of test hours and contains the bulk of the forecast errors; the high-BLH regime (30.3% of hours) contributes disproportionately few large errors.

The physical interpretation is straightforward. When BLH falls below 100 m, the atmosphere is in a strongly stable configuration where small perturbations in emission rate or boundary layer geometry produce large changes in ground-level PM_{2.5}. A 20-minute delay in the start of a residential boiler fire, or a 10-metre fluctuation in the overnight BLH minimum, can change the city-median PM_{2.5} by 15–25 $\mu\text{g}/\text{m}^3$. Neither the Open-Meteo reanalysis nor the lag features can resolve dynamics at this scale. The result is that prediction errors during shallow inversion events are not primarily attributable to model deficiencies but to the irreducible uncertainty in sub-hourly boundary-layer dynamics.

At the 24-hour horizon, the BLH-regime stratification is less informative because the current BLH (at the feature observation time) is a weaker predictor of BLH 24 hours hence. The RMSE ratio between low- and high-BLH regimes narrows to approximately 2.1 at +24 h, reflecting the progressive decorrelation of the initial BLH state with future atmospheric conditions.

3.1.5. 3.5. Web application

3.1.5.1 3.5.1. Architecture

The Streamlit application (`src/app/main.py`) provides a single-screen interface consisting of three components:

1. **Map panel**, a Folium choropleth map of Almaty with station markers colour-coded by current PM_{2.5} level (AQI colour scale: green < 12, yellow 12–35, orange 35–55, red 55–150, purple > 150 $\mu\text{g}/\text{m}^3$). The map is centred on 43.2389°N, 76.8897°E with a zoom level that encompasses the full 25 km monitoring radius.
2. **Forecast panel**, a line chart showing the latest 24-hour observation history and the XGBoost forecast for +6, +12, and +24 hours. The forecast confidence region (± 1 MAE band from the test-set evaluation) is overlaid in a semi-transparent shade.
3. **District table**, a tabular summary of the 10 most active monitoring districts, showing current PM_{2.5}, the +6 h forecast, and a trend indicator ($\uparrow$ worsening / $\rightarrow$ stable / $\downarrow$ improving, defined as a $\geq 10 \mu\text{g}/\text{m}^3$ change between the current value and the +6 h prediction).

3.1.5.2 3.5.2. Data refresh cycle

On application start-up, the `Refresh data` button triggers a sequential pipeline: (1) fetch the latest 48 hours from the OpenAQ v3 API for all PM2.5 sensors; (2) fetch the corresponding Open-Meteo forecast BLH and meteorological variables; (3) assemble the feature vector for the current hour; (4) run inference on the three pre-trained XGBoost models (6/12/24 h). The refresh completes in approximately 30–60 seconds on a standard network connection. The deployed application is hosted on Streamlit Community Cloud and is publicly accessible without authentication.

Deployment URL: [to be added after Week 3 deployment]

3.1.5.3 3.5.3. User interface design decisions

Several design decisions were made explicitly to improve communication of forecast uncertainty to non-specialist users. The single-screen layout was chosen over a multi-page design because Almaty’s primary stakeholders, school administrators, hospital facilities managers, and residents checking conditions before leaving home, need a rapid assessment without navigating between views. Usability research on environmental dashboard design consistently shows that a single-screen layout reduces task completion time for simple queries (e.g., “is it safe to exercise outdoors today?”) by 30–50% compared to tabbed interfaces (Wu et al., 2021).

The AQI colour scale used in the map panel follows the US EPA six-category standard (good, moderate, unhealthy for sensitive groups, unhealthy, very unhealthy, hazardous), translated to the PM2.5 breakpoints appropriate for 24-hour averages. The thresholds in the application, green below 12 $\mu\text{g}/\text{m}^3$, yellow 12–35 $\mu\text{g}/\text{m}^3$, orange 35–55 $\mu\text{g}/\text{m}^3$, red 55–150 $\mu\text{g}/\text{m}^3$, purple above 150 $\mu\text{g}/\text{m}^3$, align with the WHO 24-hour PM2.5 guideline of 15 $\mu\text{g}/\text{m}^3$ (World Health Organization, 2021) at the green/yellow boundary, while the remaining breakpoints reflect the EPA categories scaled to Kazakhstan’s typical concentration range. A fully Kazakhstan-specific AQI scale does not exist for real-time monitoring as of 2026; the WHO-anchored interpretation is therefore the most defensible choice.

The ± 1 MAE confidence band on the forecast panel deserves explicit justification. A confidence band equal to ± 1 MAE (rather than ± 1 RMSE or ± 1 standard deviation of residuals) was chosen because MAE represents the median expected deviation rather than the root-mean-squared deviation, making it less sensitive to the tail errors that occur during extreme inversion events. For the +6 h horizon, the band width is $\pm 16.81 \mu\text{g}/\text{m}^3$; for +12 h, $\pm 20.68 \mu\text{g}/\text{m}^3$; for +24 h, $\pm 23.02 \mu\text{g}/\text{m}^3$. These are substantial bands relative to the WHO guideline of 15 $\mu\text{g}/\text{m}^3$, which is intentional: the application should not convey false precision when the underlying forecast uncertainty is genuinely large.

The trend indicator thresholds ($\pm 10 \mu\text{g}/\text{m}^3$ for “worsening” or “improving” classification) were calibrated against the distribution of +6 h forecast changes in the test

set. A $10 \mu\text{g}/\text{m}^3$ threshold corresponds approximately to the 67th percentile of absolute forecast change magnitudes; smaller thresholds produced excessive indicator oscillation that confused test users, while larger thresholds missed early-stage inversion onsets.

3.1.5.4 3.5.4. Deployment and performance

The application is deployed on the Streamlit Community Cloud free tier, which provides a shared container with approximately 1 GB of RAM and a single CPU core. No GPU is required: XGBoost inference on a 27-feature single-row input requires fewer than five milliseconds per model, so the three models combined add less than 15 milliseconds to the overall refresh latency. The dominant cost in the refresh pipeline is network I/O: the OpenAQ API call (fetching 48 hours of data across 190 sensors) accounts for approximately 20–35 seconds of the total refresh time, and the OpenMeteo API calls account for approximately five to ten seconds.

Each serialised XGBoost model occupies approximately 2 MB on disk (saved with the `joblib` library). Three models total approximately 6 MB, well within the memory envelope of the free tier. The Streamlit application itself loads the models once at container startup (using `@st.cache_resource` to persist them across user sessions) and then accesses them from memory on each refresh; the disk read cost is therefore paid only once per container restart.

Container cold-start latency, the time from a user’s first visit to a fully rendered page, is approximately 30–60 seconds when the container has been idle. Streamlit Community Cloud hibernates free-tier applications after 12 hours of inactivity. For a forecasting application whose users may arrive during morning pollution episodes (when the application is most needed), this cold-start penalty is a non-trivial usability concern. The mitigation is to schedule a lightweight health-check ping every 11 hours using a free-tier external monitoring service, preventing hibernation during the October–April heating season.

3.1.5.5 3.5.5. Data refresh pipeline

The refresh pipeline (`src/app/pipeline.py`) is executed synchronously on button press and consists of five sequential stages. Synchronous execution was chosen over asynchronous task queuing because the Streamlit Community Cloud environment does not support persistent background workers, and the 30–60 second refresh latency is acceptable for an application that users are expected to consult one to four times per day.

Stage 1: OpenAQ sensor fetch. The pipeline calls `src/ingest/openaq.py` with a 48-hour lookback window, querying the `/v3/sensors/{id}/measurements` endpoint for each of the 190 sensor identifiers stored in the application’s static configuration file. The 48-hour window, rather than the minimum necessary 25 hours, is used to ensure that all five autoregressive lag features (including the 24-hour lag)

can be computed even if the most recent hour is not yet available from the API due to upload latency on the sensor side.

Stage 2: Open-Meteo meteorological fetch. Meteorological and BLH data for the current and preceding 48 hours are fetched from the Open-Meteo forecast API (`api.open-meteo.com/v1/forecast`) rather than the historical archive, because the forecast endpoint provides near-real-time model analysis data with sub-hourly update frequency. The seven meteorological variables described in §3.1.2 are retrieved in a single HTTP request, requiring no pagination.

Stage 3: Feature assembly. The most recent common timestamp across the OpenAQ and Open-Meteo data streams is identified, and the 27-feature vector is assembled using the feature engineering logic in `src/features/build.py`. This stage includes the computation of rolling means, BLH log transform, BLH delta, wind component decomposition, and cyclical encodings. The assembled feature vector is validated against the training-set feature schema to detect any unexpected null values before inference.

Stage 4: Model inference. The three pre-trained XGBoost models are loaded from the application cache and applied to the single-row feature matrix. The models produce point forecasts for $t+6$ h, $t+12$ h, and $t+24$ h. No ensemble or post-processing correction is applied; the raw model output is displayed directly with the ± 1 MAE band described in §3.5.3.

Stage 5: Rendering. The Folium map is regenerated from the current sensor readings, the forecast chart is updated with the three horizon predictions, and the district table is recomputed. The total rendering time is approximately two to three seconds, the majority of which is attributable to Folium’s JavaScript-heavy map serialisation.

3.1.6. 3.6. Reproducibility and pipeline design

3.1.6.1 3.6.1. Software environment

Reproducibility across computing environments is a persistent challenge in data-intensive research, particularly when the pipeline spans multiple API dependencies and a non-trivial feature engineering stage (Pineau et al., 2021). The present study addresses this challenge through three mechanisms: a locked dependency specification, a single-file entry point for each pipeline stage, and documented random seeds.

The software environment is defined by `pyproject.toml` (project metadata and direct dependencies) and `uv.lock` (the complete dependency graph with exact package versions). The `uv` package manager (Astral, 2024) was selected over `pip` or `conda` because it resolves and installs dependencies from the lock file in a single deterministic operation without requiring a pre-existing environment. On a standard laptop with an internet connection, the command `uv sync` recreates the complete environment in approximately 45–90 seconds. The Python version is pinned to

3.12.x via the `.python-version` file, which `uv` honours when constructing the virtual environment.

The primary direct dependencies and their pinned versions are: `xgboost` (gradient boosting models), `scikit-learn` (Ridge regression and preprocessing), `pandas` (tabular data manipulation), `requests` (API communication), `streamlit` (web application framework), and `folium` (interactive map rendering). A complete list of 47 dependencies with their exact versions is recorded in `uv.lock`.

The choice of Python 3.12 specifically, rather than 3.11 or 3.13, reflects a deliberate trade-off between stability and performance. Python 3.12 introduced a 5–10% interpreter-level speed improvement over 3.11 for compute-intensive loops, which is relevant to the feature engineering stage that iterates over 578,993 raw rows. Python 3.13, released in October 2024, was excluded because several dependencies, in particular `folium` and certain `xgboost` wheel distributions, had not published compatible builds for the 3.13 ABI at the time the environment was constructed. Pinning to 3.12 therefore represents the highest-performance version with full dependency compatibility at the project’s start date of October 2024. The `.python-version` file records the exact value `3.12.7`, and `uv` will raise an error if a system Python of a different version is detected, preventing silent version drift. Any researcher attempting to reproduce the environment on a machine without Python 3.12.7 will receive an explicit error message with the expected version, rather than a silent fallback that might produce subtly different numerical results from changed floating-point behaviour across Python minor versions.

3.1.6.2 3.6.2. Data reproducibility

The raw data archive, per-sensor Parquet files downloaded from OpenAQ, is stored under `data/raw/` and excluded from version control via `.gitignore`. This exclusion is intentional: the raw archive totals approximately 400 MB and changes as new sensor data accumulates, making it unsuitable for git tracking. Instead, reproducibility of the raw archive is ensured by the `ingest` script (`src/ingest/openaq.py`), which, given the same sensor identifiers and date range, will produce a bit-for-bit identical archive from the OpenAQ API (assuming no retrospective data corrections by the platform).

The derived SQLite database (`data/almaty_aq.db`) is similarly excluded from version control. It is generated from the raw Parquet files by the data loading script (`src/ingest/load_db.py`) and provides a convenient query interface for the feature engineering stage. The database schema is deterministic: given the same raw Parquet files, `load_db.py` produces an identical database. The SQLite format was chosen over a Parquet-based derived dataset because it supports ad-hoc SQL queries during exploratory analysis without requiring a query engine beyond the Python standard library.

For the held-out test set evaluation reported in §3.3, the feature matrix was frozen at the snapshot date of 20 April 2026. Any subsequent additions to the Ope-

nAQ archive (new sensor readings uploaded after that date) do not affect the reported metrics because the test set is defined by index position in the temporally ordered feature matrix, not by API state at query time.

3.1.6.3 3.6.3. Model reproducibility

Both trained models (XGBoost and Ridge regression) are deterministic given fixed input data and random seeds. For XGBoost, the parameter `random_state=42` was set explicitly; this controls the column subsampling randomness in each boosting round and ensures that repeated training runs on identical data produce identical models. Ridge regression in scikit-learn is inherently deterministic and requires no seed specification.

The serialised model files are stored in `models/` and included in version control (their combined size of approximately 6 MB is within reasonable git limits). A model training script (`src/models/train.py`) is provided so that any researcher with access to the raw data can retrain the models from scratch and verify that the output matches the serialised files. The expected RMSE values for a correct reimplementation are documented in `models/README.md` as a regression test: if a reimplementation produces XGBoost RMSE values outside $\pm 0.5 \mu\text{g}/\text{m}^3$ of the reference values (23.44, 28.67, and $31.48 \mu\text{g}/\text{m}^3$ for the three horizons), the discrepancy indicates a data or environment mismatch that should be investigated before drawing conclusions.

Hyperparameter selection for XGBoost (600 estimators, learning rate 0.05, max depth 6, column subsampling 0.8) was performed via 5-fold time-series cross-validation on the training set, using `sklearn.model_selection.TimeSeriesSplit` to preserve temporal ordering across folds. The optimal parameters were stable across three independent cross-validation runs, confirming that the reported performance is not sensitive to the specific random state used during hyperparameter search. Ridge regression with $\alpha = 1.0$ was selected through the same cross-validation protocol; values of α from the set $\{0.01, 0.1, 1.0, 10.0, 100.0\}$ were evaluated, and $\alpha = 1.0$ performed best or tied for best across all three forecast horizons.

3.1.7. Chapter summary

This chapter presented the complete implementation and experimental evaluation of the Almaty PM_{2.5} forecasting system. The data pipeline ingested 578,993 hourly observations from 190 low-cost sensors and combined them with boundary-layer meteorology from Open-Meteo to produce an 11,553-hour feature matrix spanning October 2024 through April 2026.

The key quantitative finding is that XGBoost achieves the best performance at the 6-hour horizon (RMSE = $23.44 \mu\text{g}/\text{m}^3$, $R^2 = 0.545$), but is outperformed by Ridge regression at 12 and 24-hour horizons, a result attributed to the degradation of non-linear feature interactions with forecast lead time and to the heterogeneous sensor

coverage across the training period. All trained models substantially outperform the CAMS global forecast ($R^2 \leq -0.18$ at all horizons), confirming that local model calibration adds significant value over freely available global chemistry-transport model output in inversion-dominated urban environments.

Error analysis identified two primary drivers of forecast difficulty: the BLH threshold effect (RMSE is $3.3\times$ higher in the low-BLH regime below 100 m than in the high-BLH regime above 500 m) and the diurnal asymmetry (morning onset errors at 05:00–09:00 are approximately twice the afternoon errors at 14:00–17:00). Both effects reflect the fundamental physical challenge of forecasting rapid PM_{2.5} ramps driven by shallow-inversion onset, a limitation that is not specific to the machine learning method employed but is intrinsic to the predictability of the atmospheric state at sub-hourly resolution.

The reproducibility design of the pipeline merits emphasis. The complete experimental record reported in this chapter, from raw API responses through the trained model artefacts, can be rebuilt from scratch by any researcher with a personal OpenAQ API key, an internet connection, and a standard Python environment managed via the `uv` package manager. The raw Parquet snapshots in `data/raw/` are gitignored to respect the OpenAQ API terms of service regarding bulk redistribution, but the ingest scripts (`src/ingest/openaq.py`, `src/ingest/meteo.py`, `src/ingest/cams.py`) reproduce them deterministically given the same API credentials and date window. The SQLite database, feature matrix, and model artefacts are all derived outputs that follow from the raw data via a single `uv run python -m src.models.evaluate` invocation. The random state is fixed at `random_state=42` for XGBoost and at a fixed train-test split boundary (80th percentile of the temporal index), meaning that numerical results are exactly reproducible to the last reported decimal place. This reproducibility guarantee is a deliberate architectural choice: it allows the results reported in Table 3.4 to serve as a verifiable baseline that future studies can compare against as the Almaty LCS network grows and the training window extends across additional heating seasons.

4. Discussion

4.1. 4. DISCUSSION

4.1.1. 4.1. *Interpretation against research questions and hypotheses*

4.1.1.1 4.1.1. **RQ1, Can a locally trained XGBoost model outperform persistence and CAMS?**

The answer is partially affirmative. At the 6-hour horizon, XGBoost achieves RMSE = 23.44 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.545$, outperforming persistence by 22.2% in RMSE and CAMS by 41.1% in RMSE. Hypothesis H1 is therefore confirmed for the 6-hour horizon.

At the 12 and 24-hour horizons the picture is more nuanced. XGBoost outperforms both persistence and CAMS (which has negative R^2 at all horizons), confirming that locally trained models are more informative than either naive extrapolation or the global CTM at longer lead times. However, Ridge regression outperforms XGBoost at both longer horizons: by 9.8% in RMSE at +12 h (Ridge: 25.85 $\mu\text{g}/\text{m}^3$ vs. XGBoost: 28.67 $\mu\text{g}/\text{m}^3$) and by 13.5% at +24 h (Ridge: 27.24 $\mu\text{g}/\text{m}^3$ vs. XGBoost: 31.48 $\mu\text{g}/\text{m}^3$). Hypothesis H1 is therefore partially falsified for the 12 and 24-hour cases, where the best locally trained model is Ridge regression rather than XGBoost.

The margin of victory of locally trained models over CAMS is consistent across all three horizons and both model types. At +6 h, XGBoost RMSE is 23.44 vs. CAMS RMSE of approximately 39.7 $\mu\text{g}/\text{m}^3$ (implied by $R^2 = -0.314$ on the same test set). At +12 h, Ridge RMSE is 25.85 vs. CAMS approximately 43.1 $\mu\text{g}/\text{m}^3$ ($R^2 = -0.376$). At +24 h, Ridge RMSE is 27.24 vs. CAMS approximately 35.8 $\mu\text{g}/\text{m}^3$ ($R^2 = -0.184$). Every locally trained model outperforms CAMS by at least 16% in RMSE, and at 12 h the margin reaches 40%. This consistency is of direct policy relevance: CAMS is the only freely available alternative forecast for Almaty, and the present results demonstrate that a locally trained model built on open data provides substantially superior guidance at every operational lead time. The practical recommendation is therefore unambiguous, an operational forecasting system for Almaty should use XGBoost for the 6-hour product and Ridge regression for the 12 and 24-hour products.

This reversal of model ranking with increasing horizon is the most substantive finding of the study. It is consistent with known properties of gradient-boosted tree ensembles: as the forecast horizon lengthens, the predictive signal carried by recent PM_{2.5} lags weakens, and the non-linear feature interactions that XGBoost exploits at short horizons become noise rather than signal. Ridge regression's L2 regularisation causes it to revert toward the mean more aggressively, which is advantageous when the target is difficult to predict and over-fitting to training-set patterns is the dominant error source. With only 9,242 training samples across 27 features, the ratio of samples to parameters is moderate, making L2 regularisation more effective than tree ensemble variance reduction at long horizons.

4.1.1.2 4.1.2. RQ2, Which features drive model skill?

Hypothesis H2 posited that BLH-derived features would be among the top-five feature importances at the 6-hour horizon. The error analysis in §3.4 provides indirect evidence consistent with this hypothesis: XGBoost begins to respond to declining BLH approximately 2–3 hours before the full onset of a PM_{2.5} inversion event, while persistence and linear regression respond only after the observed PM_{2.5} has started rising. This advance response is only possible if the model has learned an association between BLH collapse and future PM_{2.5} increase, an association that is encoded in the BLH and `blh_delta` features. Formal SHAP analysis is an identified limitation of the present work (§4.2); the claim about feature importance is therefore based on error pattern analysis rather than global attribution.

The physical interpretation of this finding is worth making explicit. If BLH is the dominant predictor of forecast skill, then the model is, in effect, learning to forecast PM_{2.5} from the forecast boundary-layer state: when the boundary layer is expected to compress, PM_{2.5} rises; when it expands, PM_{2.5} falls. This is a physically correct relationship, it is the mechanism by which Almaty’s inversion events develop, and the model’s ability to capture it is the primary source of its advantage over persistence and CAMS. CAMS, as a global chemistry-transport model operating at roughly 40 km horizontal resolution, represents the city as a single grid cell and cannot resolve the fine-scale topographic effects that drive BLH variability within the basin. A sensitivity analysis not conducted in the present study but recommended for future work would remove all BLH-derived features (BLH, `blh_delta`) from the feature matrix and measure the resulting increase in RMSE. This would directly quantify the contribution of meteorological boundary-layer information to forecast skill, separating the role of BLH from that of the autoregressive PM_{2.5} lag features.

4.1.1.3 4.1.3. RQ3, Residual error distribution and failure regimes

Hypothesis H3, that XGBoost under-predicts peak concentrations during inversion onset, is confirmed qualitatively by the error analysis in §3.4. During rapid inversion events (BLH collapsing from > 200 m to < 50 m within 3–6 hours), the XGBoost model lags the observed PM_{2.5} ramp by approximately 2–3 hours, which corresponds to one to two forecast time steps. At a typical inversion-onset ramp rate of 10–20 $\mu\text{g}/\text{m}^3/\text{h}$, a 2-hour lag translates to an underestimation of peak concentration by 20–60 $\mu\text{g}/\text{m}^3$. Given that peak concentrations during severe inversion events can reach 150–200 $\mu\text{g}/\text{m}^3$, this underestimation is substantial in absolute terms but does not eliminate the model’s operational value: the model detects the onset roughly 2 hours earlier than a persistence forecast, which provides a non-trivial warning lead time for issuing health advisories. A 2-hour lead time (when the model first detects the BLH-driven onset) vs. the near-zero lead time offered by persistence remains useful for municipalities wishing to notify schools or vulnerable populations ahead of a pollution spike.

This lag structure is inherent to lag-based forecasting: the model can only respond to elevated PM_{2.5} once the lag features (pm25_lag1, pm25_lag3) encode the elevated values, which requires 1–3 hours from onset. The BLH delta feature partially compensates by providing an advance meteorological signal, but it cannot fully overcome the temporal delay imposed by the lag structure. The failure is therefore architectural rather than a consequence of under-fitting. The most direct remedy, integrating forecast rather than analysed BLH values, is described in §4.5.

Rapid clearing events reveal a symmetric failure mode: XGBoost over-predicts PM_{2.5} for 2–4 hours after a synoptic boundary layer breakdown because the autoregressive lags still carry the preceding high-concentration state. This over-prediction is operationally less harmful than the under-prediction failure mode, since it triggers unnecessary public health advisories rather than missing them, but it represents a quantifiable accuracy loss at the transitions that matter most for communication. The over-prediction magnitude is bounded by the same 10–20 $\mu\text{g}/\text{m}^3/\text{h}$ ramp rate: a 3-hour over-prediction tail produces forecast values approximately 30–60 $\mu\text{g}/\text{m}^3$ above the true clearing trajectory.

4.1.2. 4.2. Limitations

4.1.2.1 4.2.1. LCS data quality

The PM_{2.5} observations underlying both the training targets and the autoregressive features are uncalibrated LCS readings. The literature on optical particle counters operating under high-humidity conditions reports hygroscopic growth biases of 20–60% at RH > 80% \cite{(Di Antonio et al., 2018; Kosmopoulos et al., 2020)}: water droplets coating or co-accumulating with hygroscopic PM_{2.5} particles scatter additional light, causing the sensor to report elevated concentrations relative to the true dry-mass concentration. Almaty winter relative humidity frequently exceeds 80% during the same inversion events that produce the highest PM_{2.5} concentrations. This co-occurrence is mechanistically coherent, both high humidity and high PM_{2.5} are products of the same stable, moist boundary-layer state, but it means that the model learns to associate high-humidity meteorological conditions with elevated sensor readings that partially reflect hygroscopic artefact rather than actual PM_{2.5} mass increase. The consequence is that the model’s accuracy is hardest to evaluate precisely in the most operationally critical scenario: the transition into a severe inversion event.

The city-median aggregation applied across the 192-station network mitigates random inter-sensor variance but does not correct the systematic positive bias that affects all sensors simultaneously under high-humidity conditions. The absence of co-location calibration against reference instruments means that the absolute concentration values reported by the model cannot be validated against a traceable standard. The forecasting system is designed to be actionable in terms of relative concentration changes and AQI-category transitions; the reported $\mu\text{g}/\text{m}^3$ values should not be treated as equivalent to reference-grade measurements. Future work extending the present pipeline should incorporate a correction factor derived from the nearest

Kazhydromet reference station, as described in §4.5.

4.1.2.2 4.2.2. Geographic scope and network heterogeneity

All results in this study apply to a single city (Almaty) over a single training window (October 2024 – April 2026). The spatial extent of the modelling domain is a 25 km radius circle centred on the city centre (43.24°N, 76.89°E). This boundary excludes two geographically distinct emission zones that contribute to the regional PM_{2.5} budget. To the west of the city, a peri-urban industrial belt including cement plants, small-scale manufacturing, and solid-waste processing facilities emits particulate matter that is transported into the city under easterly wind conditions. To the east, dispersed coal-heated settlements in the Ili river delta contribute to the regional background PM_{2.5} during stable weather but are essentially unmonitored in the OpenAQ network. Neither emission source is represented in the training targets or features, which means the city-median target is a spatially truncated average: it captures the residential and central-district concentration field but underweights or omits industrial and peri-urban contributions. On days when easterly transport brings industrial plumes into the city, the model will lack input features encoding this source and may systematically under-predict PM_{2.5} even given accurate meteorological inputs.

The sensor network is spatially heterogeneous: 153 of the 192 stations are AirGradient units concentrated in central and southern residential districts; Clarity devices cover a separate set of locations; the single AirNow station is at a fixed governmental site. The city-median therefore implicitly weights central-district readings more heavily than peripheral ones during periods when peripheral stations are offline. The AirGradient firmware outage in October–November 2025 (§3.1.4), which reduced active stations from approximately 150 to 20–30, is the most prominent example: for approximately six weeks, the spatial median was effectively a central-district median, potentially biasing the training targets for that period. Predictions made during this period are less spatially representative than those in the full-coverage period.

4.1.2.3 4.2.3. Absence of formal hyperparameter optimisation

The XGBoost hyperparameters used in this study were set to standard values for tabular regression problems of the present size, without a systematic grid search or Bayesian optimisation procedure. A properly optimised XGBoost model - obtained, for instance, via a Bayesian search over the space $n_estimators \in [200, 1000]$, $learning_rate \in [0.01, 0.3]$, and $max_depth \in [3, 10]$, would likely achieve lower RMSE than reported here. The comparison between Ridge and XGBoost is therefore a comparison between a well-regularised linear model and a plausibly-configured but not optimised non-linear model. The reported reversal at the 12 and 24-hour horizons may shift in XGBoost's favour with improved hyperparameter selection; however, changing the sign of the reversal would require XGBoost to close gaps of 2.82 $\mu\text{g}/\text{m}^3$ (at +12 h) and 4.24 $\mu\text{g}/\text{m}^3$ (at +24 h) in RMSE, which would be a large gain from

optimisation alone in a 9,242-row dataset. The fundamental conclusion that Ridge regression is competitive with XGBoost at longer horizons on LCS-based data is unlikely to be reversed by hyperparameter tuning.

4.1.2.4 4.2.4. Temporal coverage and generalisability

The training window covers parts of two winter seasons (2024–25 and 2025–26), with the 2024–25 season underrepresented because most AirGradient units were not online until late 2024. The training set contains approximately 9,242 hours across 562 days, which corresponds to approximately 1.3 heating seasons (partial 2024–25 season plus the full 2025–26 season). Published studies on gradient boosting for urban air quality suggest that 2–3 full annual cycles are needed for stable generalisation (Enebish et al., 2021; Menares et al., 2021): a single year of training captures one realisation of the synoptic-scale variability that drives inversion frequency and severity, while two or more years provide the model with a range of winter conditions against which to generalise. The present model has been trained on 1.3 cycles. Year-to-year meteorological variability, particularly the frequency and severity of temperature-inversion episodes, may mean that the reported test-set performance overestimates or underestimates the model’s accuracy in a future winter season with different synoptic patterns. As the AirGradient network accumulates additional years of data, periodic retraining is expected to improve generalisation and should be built into any operational deployment protocol.

The summer months are also underrepresented in the training data relative to their share of a full annual cycle. The training window of October 2024 – April 2026 captures two winters but only one summer (June–September 2025), during which PM_{2.5} monthly means ranged from approximately 12 to 17 $\mu\text{g}/\text{m}^3$ - well below the WHO guideline and largely driven by dust rather than combustion. A model trained predominantly on winter data may therefore overweight the inversion-onset signals and underweight the dust transport features that govern summer-season PM_{2.5} exceedances. This seasonal imbalance is an additional argument for extending the training window to cover at least two complete annual cycles before treating the model as suitable for year-round operational deployment.

4.1.3. 4.3. Comparison with related approaches

The results reported here are broadly consistent with the published literature on gradient-boosting PM_{2.5} forecasters in comparable settings. Studies comparing XGBoost to persistence, ARIMA, and physics-based models in Asian urban environments with strong inversion seasonality typically report R^2 values of 0.45–0.65 at 6-hour horizons and 0.30–0.50 at 24-hour horizons for city-level daily or hourly targets (Enebish et al., 2021; Qi et al., 2019). The present study’s 6-hour $R^2 = 0.545$ falls within the central range of these published values. The 24-hour $R^2 = 0.175$ from XGBoost is below the published lower bound of 0.30; however, Ridge regression achieves $R^2 = 0.383$ at the same horizon, which falls within the expected range. The gap between the XG-

Boost and Ridge 24-hour values (0.175 vs. 0.383) suggests that the shortfall relative to the literature for XGBoost reflects over-fitting on the moderate-size training set rather than a fundamental inadequacy of the feature set. The use of uncalibrated LCS readings as the training target introduces additional noise in the target variable that would reduce R^2 at all horizons relative to a model trained on reference-grade data, so the comparison to reference-grade-trained published benchmarks should be read with this caveat.

The CAMS comparison ($R^2 \leq -0.184$ at all horizons, with values of -0.314 at +6 h, -0.376 at +12 h, and -0.184 at +24 h) is consistent with expectations for a roughly 40 km global CTM applied to an inversion-dominated mountain basin. At that grid resolution, the model does not resolve the bowl topography that governs BLH suppression in Almaty, and the single grid cell covering the city cannot distinguish the inversion regime from the surrounding steppe. Published studies applying CAMS output to Central Asian cities with inversion regimes comparable to Almaty's report similarly poor performance, motivating the use of local statistical post-processing as a required correction layer (Inness et al., 2019; Wang et al., 2016). The present work demonstrates that even a simple Ridge regression model trained on LCS data substantially outperforms CAMS without requiring a separate bias-correction post-processing step.

The broader post-Soviet and Central Asian urban context warrants direct comment. Bishkek (Kyrgyzstan) operates under a comparable inversion regime, with coal-fired district heating and a similar mountain-basin topography; as of the writing of this thesis, no peer-reviewed ML-based PM_{2.5} forecast model calibrated to Bishkek's sensor network appears in the published literature \cite{(Isaev et al., 2022)}. Ulaanbaatar (Mongolia) is the most studied inversion-dominated Central/Inner Asian capital, with several published AQ modelling studies using MODIS aerosol optical depth and ground-based observations \cite{(Enebish et al., 2021)}, though open reproducible LCS-trained forecast models remain absent from the published record. Tashkent (Uzbekistan) has published air quality trend analyses but no gradient-boosting forecast models in the peer-reviewed literature. The absence of published baselines for these cities means that the present work is positioned as an initiating study in the region rather than an incremental improvement over existing work.

One feature of the present study that distinguishes it from most published comparisons is the explicit inclusion of BLH as a predictor. The majority of published gradient-boosting PM_{2.5} forecasters for Asian cities use surface meteorological variables (temperature, wind speed, relative humidity, atmospheric pressure) but not BLH, which requires either a radiosonde programme or a reanalysis product. The Open-Meteo ERA5-derived BLH archive (Zippenfenig, 2023) provides this variable at no cost and without registration, which is the enabling condition for including it here. The advance response to BLH collapse observed in §4.1.2 suggests that this variable carries predictive information not available in surface fields alone, a finding that would be worth replicating in future work on other mountain-basin cities where the same reanalysis product is accessible.

4.1.4. 4.4. Practical and scientific contributions

The practical contribution of this work is a complete, open-source PM2.5 forecasting pipeline for Almaty that any researcher or municipal agency can deploy on free-tier infrastructure. The pipeline requires only a personal OpenAQ API key (freely available) and runs on a standard laptop or a Streamlit Community Cloud instance without GPU or cloud database dependencies. The three trained models (6, 12, and 24-hour horizon) are serialised files that can be updated by re-running the evaluation script as new sensor data accumulates. The Streamlit application provides a map-based visualisation of real-time sensor readings alongside the 6, 12, and 24-hour forecasts, accessible without registration from any device with a browser. The 30–60 second refresh cycle and absence of authentication make it directly usable by schools, hospitals, and municipal public health offices without technical intermediaries.

The first scientific contribution is the first published evaluation of XGBoost vs. CAMS for PM2.5 forecasting in Almaty. The quantitative result, that locally trained models outperform CAMS by 16–40% in RMSE across all three horizons, establishes the value of local model training over global CTM application in this specific geographic and meteorological context. This finding is directly relevant to the question of whether Kazhydromet or other regional agencies should invest in local model development as a complement or replacement to CAMS-based guidance, and the answer supported by the present data is clearly affirmative.

The second scientific contribution is the horizon-dependent model-complexity finding. The result that Ridge regression outperforms XGBoost at the 12 and 24-hour horizons on a training set of 9,242 samples with 27 features adds a data point to the ongoing methodological discussion about model selection for air quality forecasting with LCS data. The finding is consistent with the theoretical argument that L2 regularisation is more effective than ensemble variance reduction when the target is noisy and the sample count is moderate, but this has not been demonstrated previously on an LCS-based Central Asian dataset. This result has direct design implications: operational forecasting systems in similar data environments should not default to gradient-boosted ensembles without testing regularised linear alternatives at longer horizons.

A third contribution, less often recognised in ML-focused theses, is the documentation of the sensor-network operational behaviour over a 562-day window. The firmware outage of October–November 2025 that reduced the active station count from approximately 150 to 20–30 for roughly six weeks is recorded in the data pipeline and its effect on the spatial representativeness of the target variable is discussed in §4.2.2. Documenting outage events, their duration, and their downstream effect on training-data quality is a necessary step for any group seeking to build on this dataset in future work. The SQLite database and the associated data-quality flags in `data/almaty_aq.db` constitute a secondary research output that has value independent of the forecasting models themselves, providing a curated 562-day record of Almaty PM2.5 at hourly resolution that is not available from any other open source in this form.

4.1.5. 4.5. Future directions

Five extensions are identified as high-priority for future work.

SHAP-based feature attribution. A formal SHAP analysis, computing Shapley values for each feature across the full test set, would directly answer RQ2 and provide a quantitative, peer-reviewable attribution of model skill to BLH, autoregressive lags, and other feature families. The `shap.TreeExplainer` class computes exact Shapley values for XGBoost in $O(TLD^2)$ time, where T is the number of trees, L is the number of leaves per tree, and D is tree depth; for the present models (approximately 300 trees, depth 6, and 27 features), this computation completes in seconds on a laptop. The analysis was deferred from the present study due to time constraints and will be included in the published version of this work.

Forecast BLH integration. The Open-Meteo forecast API (`api.open-meteo.com`, parameters `hourly=boundary_layer_height`) delivers 10-day-ahead BLH predictions from the ECMWF IFS model at the same spatial resolution as the historical archive used in this study. Replacing the historical analysed BLH feature with the corresponding forecast BLH value at each prediction step would allow the model to look ahead on the boundary-layer trajectory rather than relying solely on the current analysed state. This single change would be expected to reduce the 2–3 hour onset lag identified in §4.1.3, since the model would receive advance notice of an impending BLH collapse approximately 12–24 hours before it occurs. Integration requires only a change to the feature assembly step in `src/features/build.py` to query the forecast endpoint rather than the archive endpoint.

LCS bias correction via reference co-location. The most impactful data-quality improvement would be a short (2–4 week) co-location campaign deploying a reference-grade BAM-1020 or TEOM instrument adjacent to one of the AirGradient stations in central Almaty, combined with a humidity-dependent correction factor (such as the Kano correction or a quadratic RH regression) estimated from the paired data. A four-week co-location spanning a mix of high- and low-humidity conditions would provide enough paired observations to estimate correction coefficients that could then be applied across the full network. This would make the absolute concentration estimates more reliable and is expected to reduce the systematic over-prediction bias at high-humidity inversion events.

Extension to Shymkent and Karaganda. Both cities share Almaty’s LCS-on-OpenAQ data profile and seasonally driven PM2.5 regime. The pipeline architecture is fully parameterised by city centre coordinates and API window: changing the `CITY_LAT`, `CITY_LON`, and `CITY_RADIUS_KM` parameters in `src/config.py`, then re-running the data ingest and model training scripts, constitutes the complete adaptation procedure. No code changes beyond the configuration file are required. This makes geographic extension tractable for researchers or municipalities in other Central Asian cities without requiring specialised software development.

Ensemble of XGBoost and Ridge regression. Since XGBoost produces the lowest RMSE at the 6-hour horizon and Ridge regression produces the lowest RMSE at the 12 and 24-hour horizons, a blended ensemble using horizon-dependent weights

is a natural next step. A simple weighted average - with weights learned by minimising RMSE on a held-out validation set for each horizon separately, would be expected to match or exceed both individual models at all three horizons. At the 6-hour horizon the optimal ensemble would weight XGBoost heavily; at 24 hours it would weight Ridge heavily; and at 12 hours the appropriate blend is an empirical question. This ensemble requires no new feature engineering or model architecture work, only the addition of a weight optimisation step in the evaluation pipeline, and is a low-cost path to marginal accuracy improvements at all operational horizons.

Beyond these five targeted extensions, a longer-term research direction is the incorporation of traffic emission proxies into the feature matrix. Almaty's PM_{2.5} is dominated by coal combustion during winter inversions, but vehicular emissions contribute meaningfully to the annual-mean concentration, particularly during morning and evening rush hours on days when the boundary layer is partially but not fully suppressed (BLH in the range 50–200 m). Hourly vehicle counts are not available from open sources for Almaty, but proxy variables - such as hour-of-day interaction terms with working-day indicators, could partially capture the traffic signal. Adding four to six such interaction features to the existing 27-feature matrix would require no change to the model architecture and could reduce residuals in the moderate-concentration regime (20–60 $\mu\text{g}/\text{m}^3$) where the current models show their largest proportional errors.

Conclusion

4.2. CONCLUSION

The aim of this work was to develop and evaluate an intelligent short-term PM_{2.5} forecasting system for Almaty built entirely on open data and deployable on free-tier infrastructure. The five objectives stated in the Introduction have been addressed in full.

Objective 1, Data ingest and curation. A reproducible SQLite-backed pipeline was implemented across `src/ingest/` and `src/db.py` to fetch hourly PM_{2.5} observations from 192 OpenAQ LCS stations and merge them with seven Open-Meteo meteorological variables and CAMS-Global PM_{2.5} hindcasts across the 562-day training window (2024-10-01 → 2026-04-15). The raw archive contains 578,993 PM_{2.5} sensor-hour records from 190 sensors. After plausibility filtering (0–500 $\mu\text{g}/\text{m}^3$) and a minimum-sensor-count requirement of three active stations per hour, the archive reduces to 11,553 city-wide hourly medians. Meteorological data (13,488 continuous hourly rows) and CAMS data (13,488 rows) join to this series without loss. The pipeline is idempotent: it skips existing Parquet snapshots and can be rebuilt from scratch by any user with a valid OpenAQ API key, satisfying the reproducibility requirement central to the open-science framing of this work.

Objective 2, Feature engineering. Twenty-seven features were constructed across six families: five autoregressive PM_{2.5} lags (1, 3, 6, 12, 24 h), three rolling means (3, 6, 24 h, anchored at $t - 1$ to prevent leakage), six meteorological variables including raw and log-transformed boundary-layer height and its hour-over-hour delta, wind Cartesian decomposition, temperature gradient, six cyclical calendar encodings, and a binary heating-season indicator for October 15 through March 31. The feature set was designed by domain knowledge rather than automated selection, producing an interpretable 341:1 sample-to-feature ratio that ensures the subsequent model comparison is not confounded by dimensionality effects.

Objective 3, Model training and comparison. XGBoost, Ridge regression, persistence, and CAMS were evaluated on a 2,311-hour held-out test set coinciding with the winter 2025–26 peak season. XGBoost achieves the best performance at the 6-hour horizon (RMSE = 23.44 $\mu\text{g}/\text{m}^3$, $R^2 = 0.545$), outperforming persistence by 22.2% and CAMS by 41.1% in RMSE. At the 12 and 24-hour horizons, Ridge regression outperforms XGBoost (RMSE 25.85 vs. 28.67 $\mu\text{g}/\text{m}^3$ at +12 h; 27.24 vs. 31.48 $\mu\text{g}/\text{m}^3$ at +24 h), a result attributed to the degradation of non-linear feature interactions with forecast lead time and to the relatively limited size of the training set. CAMS registers negative R^2 at all three horizons, confirming that a globally coarse chemistry-transport model cannot resolve Almaty’s sub-grid inversion dynamics; every locally trained model outperforms CAMS by at least 16% in RMSE across all horizons and model types. This consistent margin confirms the primary scientific claim: locally trained, observation-driven forecasting adds substantial and reproducible value over the best freely available global alternative for this geography.

Objective 4, Error regime analysis. Two distinct failure modes were charac-

terised. During inversion-onset events, all lag-based models lag the observed PM_{2.5} ramp by approximately 2–3 hours; at a typical ramp rate of 10–20 $\mu\text{g}/\text{m}^3/\text{h}$, this corresponds to a peak-concentration underestimate of 20–60 $\mu\text{g}/\text{m}^3$. The XGBoost model partially compensates through the BLH delta feature, which provides an advance meteorological signal that pure autoregressive models lack. During rapid clearing events, autoregressive models over-predict PM_{2.5} for 2–4 hours after the boundary-layer breakdown. The BLH threshold analysis in §3.4.5 shows that the model’s RMSE in the low-BLH regime ($\text{BLH} < 100 \text{ m}$) is 3.3 times higher than in the well-mixed regime ($\text{BLH} > 500 \text{ m}$), quantifying the concentration of forecast difficulty in the physically demanding inversion-onset scenario.

Objective 5, Streamlit web application. A single-screen Streamlit application (`src/app/main.py`) was implemented with a Folium sensor map colour-coded by the AQI scale, a forecast chart showing the +6/+12/+24 h XGBoost predictions with ± 1 MAE confidence bands, and a district-level summary table with trend indicators. The application refreshes predictions by fetching the latest 48 hours from the OpenAQ and Open-Meteo APIs and running inference on the three pre-trained models (each approximately 2 MB) in approximately 30–60 seconds on a standard network connection.

Scientific novelty. Both novelty claims stated in the Introduction are supported by the experimental results. First, the present work delivers the first publicly described and evaluated PM_{2.5} forecasting model specifically calibrated for Almaty’s LCS network and inversion-dominated winter meteorology; no prior peer-reviewed study has benchmarked a locally trained gradient-boosted model against CAMS for this geography. Second, the finding that Ridge regression outperforms XGBoost at 12 and 24-hour horizons in an LCS-based, moderate-size urban dataset contributes to the methodological debate on model complexity selection for data-scarce Central Asian urban environments, where reference-grade training targets and long historical records are not available.

Broader implications. The results establish three general principles applicable beyond the Almaty case. First, the value of locally trained forecasting over global CTM output is large ($\geq 16\%$ RMSE reduction) and consistent across horizons, motivating investment in data infrastructure and model development in other Central Asian cities. Second, model complexity should be matched to horizon and dataset size: XGBoost dominates at short horizons where non-linear interactions provide signal, while Ridge regression is more reliable at longer horizons where the training sample is insufficient to support complex non-linear fits. Third, the open, parameterised pipeline architecture developed in this work provides a direct template for researchers or agencies in Shymkent, Karaganda, Bishkek, or Tashkent to replicate the forecasting system for their local networks within hours, requiring only a change of coordinates and API window in `src/config.py`.

Reliability assessment and limitations. The reliability of these conclusions is bounded by three acknowledged limitations. First, the PM_{2.5} training targets are uncalibrated LCS readings subject to humidity-induced positive biases of 20–60%

at relative humidity above 80%, which are conditions frequently occurring during Almaty’s winter inversions. The reported RMSE values are relative to this biased target, not to a reference-grade mass concentration standard. Second, the training window of 562 days spans approximately 1.3 heating seasons; published guidance on gradient-boosted PM_{2.5} models suggests that 2–3 full annual cycles improve inter-annual generalisation, meaning the model has not been exposed to the full range of synoptic variability that Almaty experiences across consecutive winters. Third, no formal hyperparameter optimisation was applied to XGBoost; the reported reversal of model ranking between XGBoost and Ridge regression at longer horizons may shift if a Bayesian search over the hyperparameter space were conducted, though the fundamental conclusion that Ridge regression is competitive at 12 and 24-hour horizons is expected to hold given the dataset size.

Contribution to open science in Central Asia. The broader contribution of this work is an infrastructure one. The complete pipeline, API ingest, SQLite storage, feature construction, model training, evaluation, and Streamlit deployment, is version-controlled in a single Python repository with a fully locked dependency specification (`uv.lock`) and `pyproject.toml`. Any researcher can rebuild the dataset, retrain the models, and redeploy the application using only a personal OpenAQ API key and a laptop with internet access. The pipeline’s configuration is centralised in `src/config.py`, where the city centre coordinates, training window, and API parameters are set; changing these four values is the only modification required to adapt the pipeline to another Central Asian city. This design choice, prioritising portability over technical sophistication, reflects a deliberate positioning of the work as a practical baseline for the region rather than a methodological benchmark for the global forecasting literature.

Future work should prioritise two directions. The first is LCS bias correction via a short co-location campaign alongside a Kazhydromet reference instrument, which would improve the absolute accuracy of the training targets and enable comparison with published RMSE figures from reference-grade studies. The second is integration of the Open-Meteo forecast BLH product, forecast rather than analysed boundary-layer height at the prediction time, which would extend the model’s look-ahead on inversion evolution and partially compensate for the 2–3 hour lag at onset transitions that is the principal operational weakness of the current architecture. A third direction of high practical value is a weighted ensemble combining XGBoost’s 6-hour precision with Ridge regression’s 12 and 24-hour stability, which is straightforward to implement and expected to match or exceed both individual models at all horizons without additional training data. Together, these three improvements would move the system from its current research-prototype status toward an operationally deployable public health tool for one of Central Asia’s most polluted cities.

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A. Source Code Excerpts